



**WATER +
PREVENTIVE
MAINTENANCE**

Water & Preventive Maintenance Foundation

水及预防性维护基础课程

AST Guidebook 讲师指南



**COFFEE
TECHNICIANS
PROGRAM**



**Specialty
Coffee
Association**

Water & Preventive Maintenance Foundation

AST Guidebook V1.0 (English)

水及预防性维护基础课程 讲师指南 v1.0 (中文版)

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Note: This guidebook replaces previous curriculum documents. This guidebook is only available to authorized trainers licensed in Water and Preventive Maintenance Foundation. Please do not share this document with other ASTs or learners

注意：先前课程文件已由本参考手册取代。相关内容仅供水与预防性维护基础课程培训师使用。请勿与其他讲师或学员分享。

1. General Information 一般信息

Course Information

Course Length: Minimum 9 hours

Minimum course hours are met through time spent in the following ways:

- The AST delivering live instruction in-person or through distance learning.
- The AST conducting informal and formal assessments. (Note: This does not include time spent by the learner taking the online written exam.)
- The learner engaging in activities that introduce and/or reinforce the theoretical or practical course objectives.

Prerequisites: None

课程信息

课程时间长度: 最短九小时

最短授课时数可透过以下方式达成:

- 由 AST 提供实体授课或远距教学
- 由 AST 进行非正式或正式评测 (请注意: 学员参加在线笔试的时间不列入时数)
- 学员参加理论类或实务类的入门和/或进阶活动

先修课程: 无

Written Exam Information:

Total Number of Questions on Online Written Exam: 30 (worth one point each)

Total Time Allowed for Online Written Exam: 32 minutes

Passing Score (Online Written Exam): 70%

Practical Exam Information:

Total Time Allowed for Practical Exam: 60 minutes

Passing Score (Practical Exam): 70% = 70 points correct / 100 total exam points

Learner must pass each section of the Practical Exam

笔试信息

在线笔试总题数: 30 题 (每题 1 分)

在线笔试时间限制： 32 分钟
及格分数（在线笔试）： 70%

实作测验信息

实作测验时间总长： 60 分钟
及格分数（实作测验）： 70% = 得分 70 / 总分 100
学生必需通过实务操作考试的每个部分

2. Course Description

Description

Water and Preventive Maintenance Foundation is an introductory course which covers key concepts of water treatment systems and preventive maintenance of coffee machines and certain practical skills required to safely apply this knowledge.

At the end of the course, the learner will be able to:

- Identify typical components and describe the function of common water treatment systems
- Define key water parameters and describe their potential effects on equipment function
- List water quality recommendations
- Describe expected effects of common water treatment methods on water quality
- Demonstrate measurement of incoming water to determine composition
- Demonstrate common steps for maintaining a water treatment system
- Identify typical wear components in a traditional espresso machine, batch brewer, and coffee grinder
- Describe the goals of a preventive maintenance program
- List the tools and supplies needed to perform preventive maintenance on coffee equipment
- Demonstrate simple preventive maintenance steps for a traditional espresso machine, batch brewer, and coffee grinder

2. 课程说明

说明

水与预防性维护的基础课程,是一堂入门课程,含概了水处理系统、咖啡机器的预防性维护及为了安全的实施这些知识所需的相关实务技巧

在课程结束后, 学生将能够达成:

- 辨认一般水处理系统的基本组成零件并能够描述其功能
- 定义主要水的参数读值, 并描述它们对设备功能的潜在影响
- 列出水质的建议
- 描述一般水处理方法在水质上的期望效果
- 示范如何对进水水源进行检测以了解其成分组成
- 示范一般维护水处理设备的步骤
- 辨认标准义式咖啡机,批次冲煮及咖啡磨豆机所需标准耗材
- 描述预防性维护计划的目标
- 列出实施咖啡设备的预防性维护所需要的工具及供应

3. Written Exam Question Distribution by Topic

The chart below sets forth key information regarding the online exam questions.

Question Pool: This is the number of questions per topic that are available to present to the learner during the online exam.

Questions Presented: This is the number of questions a learner will randomly receive per topic during the online exam. This number was determined by the creators for the purpose of ensuring that each section and topic of the course is weighted appropriately.

Section Weighting: Next to each section title is the percentage of the total exam represented by the questions in that section.

There are some topics/objectives that are not tested during the written or practical exam. The trainer should still address these topics/objectives as the information will be useful in future courses.

3. 按主题分类的笔试内容

有关在线测验试题的重要信息详如下表

题库：每个主题会用于学员在线测验的试题数量。

考题：每个主题实际列入施测内容（随机选出）的试题数量。该数量系由企划者决定，旨在确保课程每个章节和主题的比重恰如其分。

章节比重：每个章节名称右侧为相关考题在整份测验中的占比。部分主题/目标并未列入笔试或实作测验。由于这些信息将在未来课程中派上用场，因此培训师仍应讲授相关内容。

See grids on next two pages. 请参阅下两页的表格

Exam Sections & Topics 考试章节及主题	Question Pool 题库	Questions Presented 考题	Exam Sections & Topics 考试章节及主题	Question Pool 题库	Questions Presented 考题
1.01 Section Workplace Safety Practices Pre-Course 1.01 章节 工作场域安全实作 事前课程			1.04 Section Water Quality Recommendations 7% 1.04 章节 水质建议 7%		
1 Topic: Personal Responsibility 7.1.主题: 个人责任	Not Tested 未有考题		1 Topic: SCA Water Recommendations 1.主题:SCA 水质建议	3	1
2 Topic: Corporate & Manufacturing Oversight 2.主题: 公司及制造商的监管	Not Tested 未有考题		2 Topic: The SCA Water Chart 2.主题:SCA 水质图	3	1
3 Topic: Clothing & Personal Protection Equipment 3.主题: 衣着及个人保护设备	Not Tested 未有考题		3 Topic: Manufacturer Water Recommendations 3.主题:制造商的水质建议	Not Tested 未有考题	
4. Topic: Reducing Environmental Risks 4.主题:减少环境风险	Not Tested 未有考题		1.05 Section Water Treatment Methods 13% 1.05 章节 水处理方法 13%		
5 Topic: Moving Equipment 5.主题:移动设备	Not Tested 未有考题		1 Topic: Water Treatment Methods 1.主题: 水处理方法	Not Tested 未有考题	
6 Topic: Applying the Correct Tools 6.主题:运用正确的工具	Not Tested 未有考题		2 Topic: Effects of Carbon Filtration 2.主题: 活性炭滤芯的功效	3	1
7 Topic: Mental Preparedness 7.主题: 心理准备	Not Tested 未有考题		3 Topic: Effects of Water Softening 3.主题: 水质软化的功效	3	1
8 Topic: Safe Workshop Practices 8.主题: 安全工作场域的实务	Not Tested 未有考题		4 Topic: Effects of Reverse Osmosis Filtration 4.主题: 逆渗透过滤的功效	3	1
9 Topic: Recognizing Food Safety Standards 9.主题: 认知食品安全标准	Not Tested 未有考题		5 Topic: Effects of Remineralization 5.主题:水质回填的功效	3	1
1.02 Section Water Treatment System Components 7% 1.02 章节 水处理设备零件 7%			1.06 Section Measuring Water Composition 10% 1.06 章节 测量水质组成 10%		
1 Topic: Sanitary Quick-change Cartridge System Components 1.主题:干净的快拆滤芯系统零件	3	1	1 Topic: Water Testing Tools 1. 主题:水质测量工具	Not Tested 未有考题	

2 Topic: Drop-in Cartridge System Components 2.主题: 置入式滤芯系统零件	Not Tested 未有考题		2 Topic: Using an Electrical Conductivity/TDS Meter 2.主题: 使用导电度计/TDS 仪	Tested on Practical 于实务测验进行考试	
3 Topic: Reverse-Osmosis System Components 3.主题: 逆渗透系统零件	3	1	3 Topic: Using a Drop-Count Titration Kit 3. 主题: 使用滴定仪	3	1
4 Topic: Tank-style System Components 4.主题:桶式系统零件	Not Tested 未有考题		4 Topic: Using a pH Meter 4. 主题:使用 pH 仪	3	1
1.03 Section Defining Key Water Parameters 13% 1.03 章节 定义主要的水参数 13%			5 Topic: Using Water Testing Strips 5.主题:使用水质检测试纸	3	1
1 Topic: Water Composition 1.主题: 水的组成	Not Tested 未有考题		6 Topic: Using the SCA Water Chart 6.主题:使用 SCA 水质图	Tested on Practical 于实务测验进行考试	
2 Topic: Total Hardness and Alkalinity 2.主题:总硬度及碱度	4	1	1.07 Section Water Treatment System Maintenance 7% 1.07 章节 水处理系统的维护 7%		
3 Topic: Temporary and Permanent Hardness 3.主题: 暂时及永久性硬度	3	1	1 Topic: Maintenance Requirements 1.主题: 维护需求	3	1
4 Topic: Total Dissolved Solids Estimation 4.主题: 总溶解固体的估算	3	1	2 Topic: Maintaining a Cartridge System 2.主题: 滤芯系统的维护	3	1
5 Topic: pH 5.主题:pH 值	Not Tested 未有考题		3 Topic: Maintaining a Drop-in Cartridge System 3.主题: 置入式滤芯系统的维护	Tested on Practical 于实务测验进行考试	
6 Topic: Other Substances in Water 6.主题: 水中的其他物质	3	1			

Exam Sections & Topics 测验章节&主题	Question Pool 题库	Questions Presented 考题	Exam Sections & Topics 测验章节&主题	Question Pool 题库	Questions Presented 考题
1.08 Section Preventive Maintenance Components 17% 1.08 章节 预防性维护零件 17%			1.11 Section Performing Maintenance 13% 1.11 章节 维护的实施 13%		
1 Topic: Espresso Machine Brewing Group Components 1. 主题:义式咖啡机冲煮头零件	3	1	1 Topic: Safety During a PM Service 1. 主题: 预防性维护服务的安全	3	1
2 Topic: Espresso Machine Steam Valve Components 2. 主题:义式咖啡机蒸气阀零件	3	1	2 Topic: Preventive Maintenance Checklist 2. 主题: 预防性维护检核表	Tested on Practical 于实务测验进行考试	

3 Topic: Espresso Machine Steam Boiler Components 3. 主题: 义式咖啡机蒸气锅炉零件	3	1	3 Topic: Espresso Machine PM Cleaning Steps 3.主题:义式咖啡机预防性保养的清洁步骤	3	1
4 Topic: Batch Brewer Components 4. 主题:滴滤式咖啡机零件	3	1	4 Topic: Espresso Machine PM Parts Replacement 4.主题:义式咖啡机预防性保养的零件更换	Tested on Practical 于实务测验进行考试	
5 Topic: Coffee Grinder Components 5. 主题:咖啡磨豆机零件	3	1	5 Topic: Espresso Machine PM Inspection Steps 5.主题: 义式咖啡机预防性保养的检查步骤	3	1
1.09 Section Maintenance Strategies 7% 1.09 章节 维护策略 7%			6 Topic: Batch Brew Preventive Maintenance 6.主题:滴滤式咖啡机的预防性维护	3	1
1 Topic: Preventive versus Reactive Maintenance 1.主题:预防性 V.S. 反应性维护	3	1	7 Topic: Espresso Grinder Preventive Maintenance 7.主题:义式磨豆机的预防性维护	Tested on Practical 于实务测验进行考试	
2 Topic: Recommended Maintenance Intervals 2. 主题:建议保养周期	3	1			
1.10 Section Preventive Maintenance Tools and Supplies 7% 1.10 章节 预防性维护的工具与零件 7%					
1 Topic: Recommended Hand Tools 1.主题: 建议手动工具	3	1			
2 Topic: Recommended Consumables 2.主题: 建议耗材	3	1	Totals	91	30

4. Course Curriculum with Corresponding Online Written Exam Questions

The course curriculum is set forth below and is divided into Sections, Topics and Objectives. In some areas of the curriculum, the creators may have revised the curriculum in order to create a more logical, level-appropriate structure. Any revisions are noted in *2. Course Description and Updates*.

All online written exam questions were developed as an assessment for a specific objective. These questions have been grouped according to topic. All questions within a topic are considered the topic “pool.” From this pool, a certain number of questions will be randomly selected and presented to the learner. If a particular topic has more than one objective, there is a possibility that the learner will not be tested on all objectives in the topic. This is due to the randomization of the questions from that topic.

4. 课程大纲与相关的在线笔试考题

课程大纲呈现如下,并区分成区段,主题及目标. 在大纲中的某些部分,为了让整体结构更富有逻辑性及程度的適切性,创作者可能会修正大纲. 任何修正会在 2. 课程描述及更新中注明.

所有的在线笔试问题会因应特定学习目标的评估而设定.这些问题会因主题而分群. 在一个主题下的所有问题被认为是这个主题的试题库. 从这个试题库中,这些问题会在在线笔试中随机的呈现给学生. 如果某一特定主题有超过一个的学习目标, 因为主题下的问题是随机呈现的,学生可能不会在每一个学习目标都接受测验.

1.01 | Section | Workplace Safety Practices

1.01|章节|工作场域安全实作 共

11 Topics 11 个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Online Written Exam Questions	Resources
1.01.01 Personal Responsibility 个人责任	A. State that workplace safety is a shared responsibility. B. State that the individual is ultimately responsible for their own personal safety. C. State that individuals share a responsibility for the safety of those around them. A 指出工作场域的安全是一个大家共同的责任 B.指出每个人需对自己的安全负起最终的责任 C.指出个人对他们周围的安全是有共同责任的	Emphasize that workplace safety is shared responsibility and requires attention from all involved, but that each technician is responsible for keeping themselves and others around them from harm. Discuss the importance of compliance and intervention to the effectiveness of policies and procedures. 强调工作场域的安全是共同的责任,并且需要所有人的参与,但每个技师都有责任要保护他们自己及周围的人使其远离危险. 讨论规范与干预对于有效执行政策与程序的重要性	Not Tested. 无测试	
1.01.02 Corporate and Manufacturing Oversight	A. List the relevant agencies that promote policies for the protection of workers. B. List the relevant agencies and	Give examples of the relevant national and regulatory bodies. Examples include the U.S. Occupational Safety and Health Administration (OSHA), the Korean Occupational Safety and Health Agency (KOSHA), the European Agency for Safety and Health at Work (EU-OSHA), the Health and Safety Executive	Not Tested. 无测试	

公司及制造商的监管	certifications that exist to regulate safety features in equipment and tools. A. 列出推动政策来保护工作人员的相关机构 B. 列出监管设备及工具安全的相关机构及认证	ministry in the UK (HSE) and the International Labor Organization (ILO). Provide information about state and local licensing for technicians performing certain duties and give references for more information. Provide information about relevant equipment certifiers such as Underwriters Laboratories (UL), Central European (CE), Engineering Test Lab (ETL), Canadian Standards Association (CSA), and others. 举例说明相关国际及独立机关. 这些例子包含了美国职业安全卫生署(OSHA), 韩国职业安全健康局(KOSHA), 欧盟职业安全卫生署(EU-OSHA), 英国卫生安全局(HSE), 国际劳工组织(ILO) 提供执行某些任务的技师所需的国家及当地的证照且给予进一步信息的相关参考 提供相关设备证照的信息,像保险实验室(UL),欧盟认证(CE),工程测试实验室(ETL),加拿大标准协会(CSA)以及其他		
1.01.03 Clothing and Personal Protection Equipment 衣着及个人保护设备	A. List common personal protective equipment that reduces the risk of workplace injury B. List common considerations when choosing workplace clothing and accessories, including jewelry A. 列出一般减少工作场域伤害风险的个人保护性设备 B. 列出在选择工作场域服装及配件,包含首饰时的一般考虑	Give examples of how certain tasks may require specific personal protection equipment for that application. Personal protection equipment that is common in most situations include: - eye protection - appropriately-sized work gloves - ear protection Technicians should pursue: - shoes that protect their toes - clothing that fits appropriately Technicians should avoid: - wearing loose clothing - wearing loose or obtrusive jewelry such as watches, bracelets or rings. - loose long hair 举例说明某些任务如何需要特定的保护设备而加以应用在多数状况如下,个人保护设备是常见的	Not Tested. 无测验	

		-眼睛保护 -适当大小的工作手套 -耳朵保护 技师应该要贯彻: -可以保护脚趾的鞋子 -大小刚好恰当的衣着 技师应该要避免: -穿着过于宽松的衣着 -着戴宽松或突兀的首饰像是手表、手链或戒指 -松松的长发		
1.01.04 Mental Readiness 心理状态	A. Recognize mental distractions that can increase safety risk. B. Recognize when further training is necessary before attempting a test or repair. A. 认知心理上的分心会增加安全的风险 B. 认知在尝试开始测试或修理前,进一步的训练是必需的	Emphasize the important role that focus plays in maintaining a safe working environment. Discuss the different types of mental distractions that can decrease focus. These may include: - fatigue - daydreaming - worries or problems in personal life - conversations with co-workers - feeling hurried by undue pressure from customer/manager Hold a discussion about recognizing when a task goes beyond abilities or level of training. 强调在维护安全的工作环境中所扮演的重要角色 讨论会降低专注的几种心理上的分心. 他们包含了: -疲劳 -白日梦 -个人生活中的问题或烦恼 -同事间的聊天 -来自于客户或管理者的不当压力而使其感到匆忙仓促	Not Tested. 无测验	
1.01.05 Chemical Hazards 化学危害	A. Recognize the safety risks posed by chemical hazards. A. 辨认化学危害所产生的安全性风险	List the types of chemical hazards that may be faced by a coffee machine technician. Examples may include: - chemicals used for cleaning - chemicals used for descaling coffee equipment - chemicals used for lubrication Display the warning language and symbols that are used in the	Not Tested. 无测验	

		country or region of the student. 列出一个咖啡机技师可能会面临不同型的化学危害 举例包含 -清洁所使用的化学物质 -对咖啡设备除垢所使用的化学物质 -润滑所使用的化学物质 展示学生的国家或地区所使用的警语及符号		
1.01.06 Fire Hazards 火灾危害	A. Recognize the safety risks that can be posed by fire hazards in a workspace. B. State the source for information about flammable and explosive material. A. 辨识在工作场域中会产生火灾危害的安全性风险 B. 指出易燃及爆炸性物质的相关信息及来源	List the types of fire hazards that may be faced by a coffee machine technician. Examples may include: - structural fires - electrical fires - chemical fires Display the warning language and symbols that are used in the country or region of the student. Discuss the importance of emergency response plans for fire suppression or evacuation. Share resources for further study. 列出一个咖啡机技师可能会面临的火灾危害类型 举例包含 -结构性火灾 -电气火灾 -化学火灾 展示学生的国家或地区所使用的警语及符号 讨论火灾发生时,灭火及疏散警急应变计划的重要性.分享资源以达到更进一步的研究	Not Tested. 无测验	
1.01.07 Moving Equipment 设备的移动	A. State best practices for safely moving large and/or heavy equipment. A 指出安全移动较大或重的设备时所采用的最佳措施	Share the considerations and best practices for moving coffee equipment. Considerations should include: - The correct, neutral posture - The removal of tripping hazards for a clear path - Developing and communicating a plan for the movement - Using the right number of people - Proper handholds Demonstrate tools used for the cartage of coffee machines 分享移动咖啡设备时的考虑与最佳措施 考虑应包含 -正确而中性的姿势	Not Tested.	https://www.osha.gov/SLTC/etools/electricalcontractors/supplemental/principles.html#lifting

		-清除障碍物以保持干净清楚的动线通道 -发展并讨论移动的计划 -正确的人力使用 -搬运时有适当的提把处 示范搬运咖啡机所使用的工具		
1.01.08 Applying Appropriate Tools 使用恰当的工具	A. State the importance of using the correct tool for the task at hand. A. 指出使用正确的工具来完成手边工作的重要性	Discuss how using the wrong tool for a task can result in damage or injury. Discuss how the dulling of a tool that is designed to be sharp can result in damage or injury. Discuss how the condition of tools in the workplace can be regulated by state or national bodies. 讨论在工作中使用错误的工具如何能导致灾难与伤害 讨论本应被设计为锐利的工具在钝掉后如何能导致灾难与伤害 讨论国家或国际组织如何规范工作场域中工具的状况	Not Tested.	
1.01.09 Shopkeeping Practices	A. List safety considerations for the workplace environment. A. 列出工作场域环境中的安全性考虑	Discuss considerations for organizing and operating a safe workplace environment. Considerations should include: - Proper lighting - Sure footing - Clear signage and markings for warnings and directions - Restriction of workstations (moveable chains, stanchions and similar.) - Restriction of unsupervised visitors - Clearly signage for emergency response - Lockout & tagout procedures - Clearly marked supplies - Clearly marker chemicals 讨论组织及运作一个安全工作场域环境的考虑 这些考虑应该要包含: -正确的灯光 -脚下安全 -清楚易懂的警告标帜及方向指示 -工作站的管制(可移动的支柱、围条及相关类似之物) -管制未受监督的访客 -紧急应变的清楚指示 -安全上锁挂牌程序	Not Tested.	

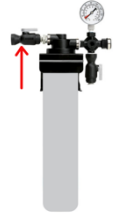
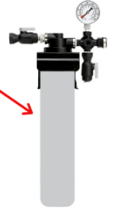
		-清楚标识的零件 -给予化学物品清楚的标识		
1.01.10 Workstation Practices 工作站的实务操作	A. List safety considerations for individual workstations. A 列出个人工作站的安全考虑	Discuss considerations for organizing and operating a safe workstation. Considerations should include: - Proper lighting - A designated place for every tool and supply - Clearly marked supplies - Appropriate height working surface - Void of spills and debris - Equipped with cleaning materials 讨论组织运作一个安全工作站的考虑 这些考虑应该要包含: -正确的灯光 -保留清楚标识的空间给每个工具及零件 -清楚标识的零件 -恰当高度的工作枱面 -保留会溅出或有碎片的空间	Not Tested. 无测试	
1.01.11 Respecting Food Safety Standards 遵守食物安全标准	A. List best practices that respect food safety standards while working in a food handling space. A. 列出在食物加工环境工作时,能达成遵守食物安全标准的最佳措施	Emphasize that technicians must be aware of the manufacturer's cleaning recommendations for each piece of equipment. Discuss the surfaces of a coffee machine that come in contact with the beverage product and the importance of cleanliness to the customer. Discuss how certain cleaning compounds, such as alkyl sulfonate, can have offensive tastes and are detectable at concentrations of 100 ppm. 强调技师必需清楚明白制造商对于设备中每一部分的清洁建议. 讨论咖啡机中与饮品有所接触的表面以及之于顾客端清洁的重要性 讨论某些清洁化合物,像是烷基磺酸酯,会有讨厌的味道,并且在浓度达到 100ppm 时是可以被察觉到的	Not Tested. 无测验	

1.02 | Section | Water Treatment System Components

1.02 | 章节 | 水处理设备零件 共

4 Topics 4 个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线笔试问题	Resources 相关资源
1.02.01 Quick-change Cartridge System Components 快拆滤芯系统零件	A. List common components of a quick-change cartridge water treatment system, including: - System head - Cartridge - Shut-off valve(s) - Pressure gage A. 列出一般快拆过滤水处理系统的组成零件,包含: -卡式滤头 -快拆卡式滤芯 -进水阀 -压力计		Question ID: 100010842451 A commonly used type of water filter system is shown in this picture. Which of the following terms best describes the system component designated by the arrow?  System head Quick-change cartridge Booster pump Shut-off valve 问题编号: 100010842451 在照片中我们可以看到一组常见的水过滤系统. 下方哪一个用词最适合用来描述箭头所标出的零件 卡式滤头 快拆卡式滤芯 加压帮浦 进水阀 Question ID: 100010842452 A commonly used type of water filter system is shown in this picture. Which of the following terms best describes the system component designated by the arrow?	

			 <u>Shut-off valve</u> System head Quick-change cartridge Booster pump 问题编号: 100010842452 在照片中我们可以看到一组常见的水过滤系统. 下方哪一个用词最适合用来描述箭头所标出的零件 <u>进水阀</u> 卡式滤头 快拆卡式滤芯 加压帮浦 Question ID: 100010842453 A commonly used type of water filter system is shown in this picture. Which of the following terms best describes the system component designated by the arrow?  <u>Quick-change cartridge</u> System head Booster pump Shut-off valve 问题编号: 100010842453 在照片中我们可以看到一组常见的水过滤系统. 下方哪一	
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			个用词最适合用来描述箭头所标出的零件 <u>快拆卡式滤芯</u> 卡式滤头 加压帮浦 进水阀	
	B. Identify common components of a quick-change cartridge water treatment system B. 办认一组快拆水处理系统的组成零件	Activity 1.02, See Appendix A - Activity Preparation Guide 活动 1.02 看附录 A 活动准备指南	This objective has an activity. 这个教学目标有活动	
1.02.02 Drop-in Cartridge System Components 置入式过滤系统零件	A. List common components of a drop-in water treatment system, including: - System housing - O-ring - Shut-off valve(s) - Cartridge A. 列出一般置入式水处理系统的零件,包含: - 系统滤水器外壳 - O 形密封硅胶环 - 进水阀 - 滤芯	Discuss various configurations of components which may be used by different manufacturers 讨论不同制造商所使用各式不同的零件配置与构造	Not tested. 无测验	
	B. Identify common components of a drop-in water treatment system B 辨认一组置入式过滤系统的组成零件	Activity 1.02, See Appendix A - Activity Preparation Guide 活动 1.02 看附录 A 活动准备指南	This objective has an activity. 这个课程目标有活动	

1.02.03 Reverse-Osmosis System Components 反渗透过滤系统零件	A. List common components of a reverse-osmosis water treatment system, including: - System head - Booster pump - Sediment filter - Carbon filter - Membrane - Blending valve - Storage tank A.列出反渗透过滤处理系统的一般零件,包含: -系统上座 -加压帮浦 -棉质滤芯 -活性炭滤芯 -过滤膜 -混水阀 -储水桶	Discuss various configurations of components which may be used by different types of reverse osmosis systems and different manufacturers 讨论不同制造商及不同的反渗透过滤系统所使用各式不同的零件配置与构造	Question ID: 100010842476 Which type of water filtration system often uses a booster pump and a membrane? <u>Reverse osmosis system</u> Water softener Carbon filter Remineralizing system 问题编号: 100010842476 哪一种型态的水过滤系统通常会用到加压帮浦和过滤膜? <u>反渗透过滤系统</u> 软水系统 活性炭滤芯 矿物质回填系统 Question ID: 100010842480 Which of the following components are NOT typically part of a reverse osmosis water treatment system? <u>Water softener cartridge</u> Carbon filter Booster pump Membrane 问题编号: 100010842480 下方哪一个零件并未在经典的反渗透过滤水处理系统中? <u>软水滤芯</u> 活性炭滤芯 加压帮浦 过滤膜 Question ID: 100010842485 Which of the following components is missing from this reverse osmosis system: 1. Booster pump 2. System head 3. Carbon Filter	
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			4. _____ 5. Storage Tank Membrane Softener cartridge Polyphosphate cartridge Decarbonizing cartridge 问题编号: 100010842485 下方哪一个零件在逆渗透过滤系统中消失了: 1. 加压帮浦 2. 系统上座 3. 活性炭滤芯 4. 5. 储水桶 过滤膜 软水滤水器外壳 多磷酸盐滤芯 脱碳滤芯	
1.02.04 Tank-style System Components 桶式系统零件	Identify a tank-style water treatment system 辨认桶式水处理系统		Not tested. 无测验	

1.03 | Section | Defining Key Water Parameters

1.03 | 章节 | 定义主要水的参数

6 Topics 六个主题

Topic 主题	Objectives 目标	AST Notes 讲师重要	Written Online Exam Questions 在线笔试题目	Resources 相关资源
1.03.01 Water Composition 水的组成	A. List typical sources of substances commonly dissolved in water, including: - the atmosphere - contact with rock and soil - pollution of surface water - municipal water treatment process - saltwater intrusion A.列出典型一般溶解在水中物质的来源,包含: -大气 -与岩石及土壤的接触 -地表水的污染 -自来水处理设备的过程 -沿海低洼地区的盐水入侵		Not tested. 无测验	The SCA Water Quality Handbook, pg. 10 SCA 水质手册 第 10 页
1.03.02 Total Hardness and Alkalinity 总硬度及碱度	A. Define total hardness as the combined quantity of calcium and Magnesium ions dissolved in water B. Define alkalinity as the acid buffering capacity of water and concentration of carbonate and bicarbonate compounds in the water C. State that the recommended unit of measure for total hardness and alkalinity is ppm CaCO ₃ A.定义总硬度为溶解于水中钙离子	Alkalinity is sometimes called carbonate hardness AST should discuss that alkalinity differs from Alkaline - having a pH higher than 7. 碱度有时被称为碳酸盐硬度 讲师应讨论碱度与碱性并不相同 碱性为 PH 值大于 7	Question ID: 100010842659 Which of the following parameters describes the combined quantity of calcium and magnesium ions dissolved in water? <u>Total hardness</u> Permanent hardness Alkalinity Total dissolved solids 问题编号:100010842659 下面哪一个参数用以描述水中溶解钙离子与镁离子的总量? <u>总硬度</u>	The SCA Water Quality Handbook, pg. 12 SCA 水质手册 第 12 页

	与镁离子的总合 B 定义碱度为水中缓冲酸的能力,为水中碳酸根及碳酸氢根浓度		永久硬度 碱度 总溶解固体量	
			Question ID: 100010842661 Is the following statement true or false? Total hardness defines the quantity of dissolved calcium carbonate in water. <u>False</u> True 问题编号:100010842661 请问以下的论述正确与否? 总硬度定义为水中溶解碳酸钙的量值? <u>错误</u> 正确	The SCA Water Quality Handbook, pg. 12 SCA 水质手册 第 12 页
			Question ID: 100010842662 Which of the following units is commonly used to quantify total hardness? ppm CaCO ₃ German degrees Grains per Gallon <u>All of the above</u> 问题编号:100010842683 下面哪一个通常被用来作为总硬度测量的单位? PPM CaCO ₃ 德制硬度 克/加仑 <u>以上皆是</u>	The SCA Water Quality Handbook, pg. 15 SCA 水质手册 第 15 页
	D. List other common measurement units for total hardness and alkalinity, including: - German degrees - Grains per US Gallon D. 列出其他一般用来作为总硬度或		Question ID: 100010842664 Which of the following units is commonly used to quantify alkalinity? <u>ppm CaCO₃</u> μS Degrees Celsius	The SCA Water Quality Handbook, pg. 16 SCA 水质手册 第 16 页

	碱度测量的单位? -德制硬度 -克/美式加仑		pH 问题编号:100010842664 下面哪一个通常被用来作为总碱度测量的单位? <u>ppm 碳酸钙</u> μS 摄氏温度 pH	
1.03.03 Temporary and Permanent Hardness 暂时及永久性硬度	A. Define Temporary and Permanent Hardness A. 定义暂时性及永久性硬度		Question ID: 100010842665 Which of the following water parameters indicates the dissolved calcium and/or magnesium which CAN precipitate out of a sample of water when boiled? <u>Temporary hardness</u> Permanent hardness Total hardness Alkalinity 问题编号:100010842665 下面哪一个水的参数指的是在水被煮滚后,水中溶解会被沉淀析出的钙及镁? <u>暂时硬度</u> 永久硬度 总硬度 碱度 Question ID: 100010842666 Which of the following water parameters indicates the dissolved calcium and/or magnesium which CANNOT precipitate out of a sample of water when boiled? <u>Permanent hardness</u> Temporary hardness Total hardness Alkalinity 问题编号: 100010842666	The SCA Water Quality Handbook, pg. 47 SCA 水质手册 第 47 页

			下面哪一个水的参数指的是在水被煮滚后,水中溶解<u>不会</u>被沉淀析出的钙及镁? <u>永久硬度</u> 暂时硬度 总硬度 碱度 Question ID: 100010842667 Which of the following best defines temporary hardness? <u>Dissolved calcium and/or magnesium which can precipitate out of a sample of water when boiled</u> Dissolved calcium and/or magnesium which cannot precipitate out of a sample of water when boiled Dissolved calcium and/or magnesium which occur as compounds other than carbonates in water The total quantity of dissolved calcium and/or magnesium 问题编号: 100010842667 下方哪一个选项是暂时硬度的最佳定义? <u>水中溶解的钙与镁在水煮滚后会被沉淀析出</u> 水中溶解的钙与镁在水煮滚后不会被沉淀析出 水中溶解的钙与镁以其他非碳酸盐化合物的形式存在 水中溶解的钙与镁的总量	
1.03.04 Total Dissolved Solids Estimation 总溶解固体的估算	A. State that the electrical conductivity of water can be used to estimate the total dissolved solids (TDS) in water A. 指出水的导电度可用来估算水中的总溶解固体(TDS)含量	Handheld TDS meters actually measure the electrical conductivity of the sample which is then used to display TDS based on a conversion factor. 手持式的 TDS 测量仪实际上测量的是样本的导电度,以其做转换因子而成为 TDS 的概略值	Question ID: 100010842668 What physical parameter of a water sample does a handheld TDS meter measure in order to produce a TDS reading? <u>Electrical conductivity</u> Total dissolved solids pH Refractive index 问题编号: 100010842668 为了制造 TDS 的读数,手持式 TDS 测量仪以水样本中的哪一个物理参数来做测量?	Water for Coffee, pg 56-58 咖啡中的水 56-58 页

			<u>导电度</u> 总溶解固体含量 pH 折射率 Question ID: 100010842669 Is the following statement true or false? Handheld TDS meters measure the electrical conductivity of a liquid and display an estimate of the total dissolved solids in the sample. <u>True</u> False 问题编号: 100010842669 以下的论述正确与否? 手持式 TDS 测量仪测量液体的导电度并显示样本总溶解固体含量的概略值 <u>正确</u> 错误 Question ID: 100010842670 Which of these easily measured parameters is often used to estimate the total dissolved solids in a water sample? <u>Electrical conductivity</u> pH Turbidity Hardness 问题编号: 100010842670 下列这些容易测量的参数中,哪一个通常被用来估算样本的水中总固体含量 <u>导电度</u> PH 值 浊度 硬度	
1.03.05 pH pH 值	A. Define pH A. 定义 PH 值		Not tested. 无测验	

1.03.06 Other Substances in Water 其他水中物质	A. State that other substances are sometimes found in drinking water, including: - Sodium - Iron - Chlorine - Chlorides - odor-causing compounds - sediment A. 指出饮用水中有时会找到的其他物质, 包含: -钠 -铁 -氯 -有味道物质 -沉淀物		Question ID: 100010842672 Which of the following substances is sometimes found dissolved in drinking water? Sodium Iron Odor-causing compounds <u>All of the above</u> 问题编号: 100010842672 下方哪一个物质有时会被发现溶解于饮用水中 钠 铁 有味道化合物 <u>以上皆是</u> Question ID: 100010842673 Which of the following is an undesirable substance that may sometimes be found in municipal drinking water? <u>Sediment</u> Calcium Magnesium Bicarbonate 问题编号: 100010842673 下方哪一个物质有时会被发现于供水系统的饮用水中,却是一个令人讨厌的物质 <u>沉淀物</u> 钙 镁 碳酸氢根	Equilibrium in Water and Coffee, pg 86 水与咖啡的平衡
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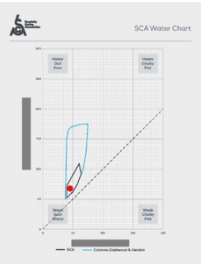
			Question ID: 100010842675 Which of the following is an undesirable substance that may sometimes be found in municipal drinking water? <u>Odor-causing compounds</u> Alcohol Magnesium Bicarbonate 问题编号: 100010842675 下列哪一个令人讨厌的物质有时会出现于供水系统的饮用水中? <u>有味道的物质</u> 酒精 镁 碳酸氢根	
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1.04 | Section | Water Quality Recommendations 1.04 | 章节 | 水质建议

3 Topics 三个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线笔试问题	Resources 相关资源
1.04.01 SCA Water Recommendations SCA 水质建议	A. Recognize that the SCA Water Quality Handbook recommends that water for coffee extraction should: - Be odor free and hygienic - Have a pH of between 6-8 - Be free of iron, lead, or other unsuitable or toxic compounds in significant amounts A. 辨识 SCA 水质手册建议咖啡萃取用水应该: -无外来的味道且卫生 -PH 值在 6-8 之间 -没有铁、铅或其他不恰当或明显量质的有毒物质	Legacy SCAA Water Quality Standard specified that Chlorine be limited to 0 mg/L and sodium limited to 10mg/L (for Superior Brew) 传统 SCAA 水质标准明确指出氯含量的限制为零,而钠含量为 10mg/L(为了更好的冲煮)	Question ID: 100010842683 Which of the following characteristics does the SCA recommend that water for coffee brewing exhibit? Odor free Hygienic Free of iron, lead, or toxic compounds <u>All of the above</u> 问题编号:100010842683 下方的哪个特质符合 SCA 建议的咖啡冲煮用水? 无外来风味 卫生 无铁、铅或其他有毒物质 <u>以上皆是</u>	SCA Water Quality Handbook, pg. 74-76 SCA 水质手册 74-76 页
			Question ID: 100010842685 What range of pH does the SCA recommend for water for coffee extraction? <u>6-8</u> 5-7 7-9 There is no recommendation 问题编号:100010842685 SCA 建议咖啡萃取用水的 PH 值范围为何? <u>6-8</u> 5-7 7-9 没有建议	

			Question ID: 100010842686 According to SCA Water Quality Handbook, which of the following substances is acceptable in moderate amounts in water for coffee brewing? <u>Calcium</u> Chlorine Iron Lead 问题编号:100010842685 根据 SCA 水质手册,在咖啡冲煮用水中,下方哪一个物质适量存在是可以接受的? <u>钙</u> 氯 铁 铅	
1.04.02 The SCA Water Chart SCA 水质图	A. Recognize that the SCA Water Chart represents recommended levels of total hardness and alkalinity for water for coffee brewing B. State that the SCA Water Chart recommendation minimizes the risk of limescale formation and corrosion in espresso machines and hot water boilers A. 辨识 SCA 水质图代表了对咖啡冲煮使用水整体硬度及碱度的建议数值 B. 指出 SCA 水质图的建议会极小化义式咖啡机及热水锅炉中水垢的形		Question ID: 100010842689 Which two (2) of the following water parameters are shown on the SCA Water Chart? <u>Total hardness</u> <u>Alkalinity</u> Total dissolved solids (TDS) Permanent hardness 问题编号: 100010842689 在下方水参数中,有哪二个出现在 SCA 的水质图中? <u>总硬度</u> <u>碱度</u> 总溶解固体含量(TDS) 永久硬度	

	成及管线的锈蚀		Question ID: 100010842691 Which two (2) of the following potential equipment risks does the SCA Water Chart's Core Zone minimize? <u>Limescale formation</u> <u>Corrosion</u> Silicate deposits Iron deposits 问题编号: 100010842691 SCA 水质图中的中心区域可以极小化下方潜在设备风险中的哪二个? <u>水垢的形成</u> <u>锈蚀</u> 硅酸盐的沉淀 铁的沉淀 Question ID: 100010842692 Which of the following is a risk of using the water with the parameters shown here on the SCA Water Chart?  <u>There is no risk</u> Limescale formation Pitting corrosion Silicate buildup 问题编号: 100010842692 使用如图所示 SCA 水质图中的水质参数,会有什么样的风险? <u>无风险</u> 结水垢 孔蚀 硅酸盐积聚沉淀	
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1.04.03 Manufacturer Water Recommendations 制造商的水建议	A. Recognize that equipment manufacturers provide recommendations for water composition, which should be followed. B. State that water containing levels of chlorides above manufacturer recommendations may result in corrosion in coffee equipment A. 应当辨识并遵守设备制商所提供的水质组成建议 B. 指出水中氯含量高于设备制造商的建议时可能会导致咖啡设备的锈蚀		Not tested. 无测试	
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1.05 | Section | Water Treatment Methods 1.05 | 章节 | 水处理的方法
5 Topics 五个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线笔试问题	Resources 相关资源
1.05.01 Water Treatment Methods 水处理的方法	A. List commonly used water treatment methods, including: - carbon filtration - sediment filtration - water softening - reverse osmosis filtration - remineralization A. 列出通常会用来进行水处理的方法,包含: -活性碳滤芯 -棉质滤芯		Not tested. 无测验	

	-水质软化 -逆渗透过滤 -矿物质回填			
1.05.02 Effects of Carbon Filtration 活性炭滤芯的功用	A. List expected changes to water composition as a result of carbon filtration B. List aspects of water composition which are not expected to be changed as a result of carbon filtration A. 列出在经过活性炭滤芯后所期待的水质组成改变 B. 列出在经过活性炭滤芯后,无法改变的水质组成部分		Question ID: 100010842696 Which of the following effects is expected if water is treated using a carbon filter? <u>Odor-causing compounds are removed</u> Significant reduction in total hardness Significant reduction in alkalinity Significant reduction in total dissolved solids 问题编号: 100010842696 如果使用活性炭滤芯作为水处理的方法,下方哪个功效是被期待的? <u>去除有味道物质</u> 在总硬度上的明显下降 在总碱度上的明显下降 在总溶解固体含量上的明显下降	
			Question ID: 100010842698 Which two (2) of the following effects are not expected if water is treated using a carbon filter? <u>Significant reduction in total hardness</u> <u>Significant reduction in alkalinity</u> Odor-causing compounds are removed Chlorine taste is removed 问题编号: 100010842698 如果使用活性炭滤芯作为水处理的方法,下方哪二个功效是不被期待的? <u>在总硬度上的明显下降</u> <u>在总碱度上的明显下降</u> 去除有味道物质 去除氯的味道	

			Question ID: 100010842699 Which of the following effects is expected if water is treated using a carbon filter? <u>Unpleasant taste is removed</u> Significant reduction in total hardness Significant reduction in alkalinity Significant reduction in total dissolved solids 问题编号: 100010842699 如果使用活性炭滤芯作为水处理的方法,下方哪个功效是被期待的? <u>去除不舒服的味道</u> 在总硬度上的明显下降 在总碱度上的明显下降 在总溶解固体含量上的明显下降	
1.05.03 Effects of Water Softening 水质软化的功效	A. List expected changes to water composition as a result of water softening B. List aspects of water composition which are not expected to be changed as a result of water softening A. 列出在经过水质软化滤芯后所期待的水质组成改变 B. 列出在经过水质软化滤芯后,无法改变的水质组成部分	The SCA Water Chart should be used to visually represent the expected changes to total hardness and alkalinity during discussions 在讨论时,SCA 水质图应该用以视觉上观察总硬度及碱度所期待的改变	Question ID: 100010842702 Which of the following effects is expected if water is treated using a water softener? <u>Significant reduction in total hardness</u> Odor-causing compounds are removed Significant reduction in alkalinity Significant reduction in total dissolved solids 问题编号: 100010842702 如果使用水质软化滤芯作为水处理的方法,下方哪个功效是被期待的? <u>在总硬度上的明显下降</u> 去除有味道物质 在总碱度上的明显下降 在总溶解固体含量上的明显下降	

			Question ID: 100010842703 Which two (2) of the following effects are NOT expected if water is treated using a water softener? <u>Significant reduction in alkalinity</u> <u>Significant reduction in total dissolved solids</u> Significant reduction in total hardness Reduced risk of limescale buildup 问题编号: 100010842703 如果使用水质软化滤芯作为水处理的方法,下方哪二个功效是不被期待的? <u>在总碱度上的明显下降</u> <u>在总溶解固体物质上的明显下降</u> 在总硬度上的明显下降 减少水垢形成的风险	
			Question ID: 100010842704 Which of the following water treatment methods is typically used to reduce only the total hardness of water? <u>Water softening</u> Remineralization Carbon filtration Reverse osmosis 问题编号: 100010842704 下方的哪一个水处理方法通常被只用来减少水中的总硬度 <u>水质软化滤芯</u> 矿物质回填滤芯 活性炭滤芯 反渗透过滤	
1.05.04 Effects of Reverse Osmosis Filtration 反渗透过滤的功效	A. List expected changes to water composition as a result of reverse osmosis filtration B. List aspects of water composition which are not expected to be	The SCA Water Chart should be used to visually represent the expected changes to total hardness and alkalinity during discussions	Question ID: 100010842705 Which of the following effects is expected if water is treated using a reverse osmosis system? Significant reduction in total hardness Significant reduction in alkalinity	

	changed as a result of reverse osmosis filtration C. State that controlled amounts of untreated water can be blended with reverse osmosis treated water prior to final filtration. Describe the expected changes to the water composition. A. 列出在经过反渗透过滤系统后所期待的水质组成改变 B. 列出在经过反渗透过滤系统后,无法改变的水质组成部分 C.指出将已比例控制的未处理水与反渗透过滤系统后的处理水加以混合,再进行最终的过滤,描述在水质组成上所期待的变化。	在讨论时,SCA 水质图应该用以在视觉上代表在总硬度及碱度所期待的变化	Significant reduction in total dissolved solids <u>All of the above</u> 问题编号:100010842705 如果使用反渗透过滤系统作为水处理的方法,下方哪个功效是被期待的? 在总硬度上的明显下降 在碱度上的明显下降 在总溶解固体含量上的明显下降 <u>以上皆是</u> Question ID: 100010842706 Which of the following effects are NOT expected if water is treated using a reverse osmosis system? <u>Increase in total dissolved solids</u> Significant reduction in alkalinity Significant reduction in total dissolved solids Significant reduction in total hardness 问题编号: 100010842706 如果使用反渗透过滤系统作为水处理的方法,下方哪个功效是不被期待的? <u>增加总溶解固体含量</u> 在碱度上的明显下降 在总溶解固体含量上的明显下降 在总硬度上的明显下降 Question ID: 100010842707 Which of the following approaches may be used to increase the mineral level of water which has been treated using a reverse osmosis system? <u>Blending in some untreated water prior to final filtration</u> Ion exchange softening Decarbonization water softening	
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			Carbon filtration 问题编号: 100010842707 下方的何种方法可用于在逆渗透过滤系统中增加水中的矿物质? <u>在最终过滤前与一些未处理的水加以混合</u> 离子交换以软化水质 脱碳水质软化 活性碳滤芯	
1.05.05 Effects of Remineralization 矿物质回填的功效	A. List expected changes to water composition as a result of remineralization B. List aspects of water composition which are not expected to be changed as a result of remineralization A. 列出在经过矿物质回填系统后所期待的水质组成改变 B. 列出在经过矿物质回填系统后,无法改变的水质组成部分		Question ID: 100010842708 Which of the following effects is expected if water is treated using remineralization? <u>Increased total hardness and alkalinity</u> Reduced total hardness and alkalinity Odor-causing compounds are removed Significant reduction in total dissolved solids 问题编号: 100010842708 如果使用矿物质回填系统作为水处理的方法,下方哪个功效是被期待的? <u>增加总硬度及碱度</u> 减少总硬度及碱度 去除有味道物质 在总溶解固体含量上的明显下降 Question ID: 100010842709 Which two (2) of the following effects are NOT expected if water is treated using remineralization? <u>Reduction in total dissolved solids</u> <u>Odor-causing compounds are removed</u> Increased total hardness Increased alkalinity 问题编号: 100010842709 如果使用矿物质回填系统作为水处理的方法,下方哪二个功效是不被期待的?	

			<u>在总溶解固体含量上的明显下降</u> <u>去除有味道物质</u> 增加总硬度 增加碱度	
			Question ID: 100010842710 Which of the following water treatment methods is typically used to increase the mineral content of water? <u>Remineralization</u> Water softening Carbon filtration Reverse osmosis 问题编号: 100010842710 下方的哪一个水处理方法通常被用来增加水中的矿物质含量? <u>矿物质回填滤芯</u> 水质软化滤芯 活性炭滤芯 反渗透过滤	

1.06 | Section | Measuring Water Composition 1.06 | 章节 | 测量水质的组成
6 Topics 六个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线考试问题	Resources 相关资源
1.06.01 Water Testing Tools 水质测量工具	A. List measurement tools commonly used for measuring water composition, including: - Electrical conductivity/TDS meters - Drop-count titration kits - pH meters - test strips B. Describe best practices for collecting and testing water samples A. 列出一般在测量水质组成所需工具,包含: -导电度计/TDS 测量仪 -滴定仪 -PH 仪 -检测试纸 B. 描述收集及检测水样本的最佳措施		Not tested. 无测验	
1.06.02 Using an Electrical Conductivity/TDS Meter 使用导电度计/TDS 测量仪	A. State limitations of electrical conductivity-based measurement of total dissolved solids A. 指出以导电度为基准来进行总溶解固体浓度测量的限制	The electrical conductivity of samples varies depending on the exact composition of the dissolved minerals as well as the temperature of the sample, so the true TDS of a sample can only be estimated using this method. Note that meters may use different conversion factors Estimated values can be useful if total hardness and alkalinity are known	Not tested. 无测验	Water for Coffee, pg 56-58

		样本的导电度会因实际溶于水中的矿物质组成及温度而有所不同,因此某一样本真正的总溶解固体浓度仅能以此方法进行估算 需注意测量仪可能会用不同的换算因子 在总硬度及碱度是可知的状态下,总溶解固体浓度的概略值是有效的		
	B. Demonstrate proper usage of an Electrical Conductivity/TDS meter B. 示范正确使用导电度计/TDS 测量仪	Activity 1.06, See Appendix A - Activity Preparation Guide 活动 1.06 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
1.06.03 Using a Drop-Count Titration Kit 使用滴定仪	A. Recognize that drop count titration is a recommended method for assessing total hardness and alkalinity of a water sample A. 辨别滴定法是测量水质样本的总硬度及碱度的建议方法		Question ID: 100010842711 Which of the following testing methods is recommended for use by a coffee equipment technician to measure the total hardness of a water sample? <u>Drop-count titration</u> TDS meter pH meter Test strips 问题编号: 100010842711 下列哪一个为建议咖啡设备技师用来测量水质样本的总硬度的测量方式? <u>滴定法</u> TDS 测量仪 PH 仪 测量试纸	

			Question ID: 100010842712 Which of the following testing methods is recommended for use by a coffee equipment technician to measure the alkalinity of a water sample? <u>Drop-count titration</u> TDS meter pH meter Test strips 问题编号: 100010842712 下列哪一个为建议咖啡设备技师用来测量水质样本碱度的测量方式? <u>滴定法</u> TDS 测量仪 PH 仪 测量试纸	
			Question ID: 100010842713 Which two (2) of these coffee brewing water parameters are often measured using the drop count titration method? <u>Total hardness</u> <u>Alkalinity</u> Total Dissolved Solids Temperature 问题编号: 100010842713 下列哪二个咖啡冲煮参数通常会用滴定法来加以测量? <u>总硬度</u> <u>碱度</u> 总溶解固体含量 温度	

	B. Demonstrate the proper use of drop-count titration to measure the total hardness and alkalinity of a water sample. B.示范正确使用滴定法来测量水质样本的总硬度和碱度	Activity 1.06, See Appendix A - Activity Preparation Guide 活动 1.06 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
1.06.04 Using a pH Meter PH 仪的使用	A. Recognize that a pH meter is a recommended method for assessing the pH of a water sample B. Describe the calibration process and storage recommendations for a pH meter A. 辨别 PH 仪为测量水质样本 PH 值的建议方法 B. 描述 PH 仪的建议校正方式及储存模式		Question ID: 100010842736 Is the following statement true or false? The electrode on a pH meter has a limited-service life and may need to be replaced periodically. True False 问题编号: 100010842736 请问下方陈述正确与否? pH 仪的电极有其有限的使用寿命且可能会需要周期性的更新 是 否 Question ID: 100010842737 Which of the following steps are recommended before using a pH meter? <u>Calibrate using an appropriate buffer solution</u> Clean thoroughly using alcohol wipes Ensure that the electrode bulb is thoroughly dry Store fully submerged in distilled water 问题编号: 100010842737 在使用 pH 仪前,下列步骤哪个是建议选项? <u>用适当的缓冲溶液进行校正</u> 用酒精彻底擦拭 确定前端电极球彻底干燥 存放时完整地浸泡于干净的蒸馏水中	

			Question ID: 100010842738 How should a pH meter be stored? <u>With a small amount of storage solution in the cap</u> Fully submerged in distilled water With a clean and thoroughly dry bulb Upside-down 问题编号: pH 仪应该如何保存呢? <u>盖中需有少量的储存溶液</u> 整支浸泡于蒸馏水中 有着干净且完全干燥的电极球 上下颠倒	
	C. Demonstrate the proper use of a pH meter to measure the pH of a water sample. C. 示范正确使用 pH 仪来测量水质样本的 pH 值	Activity 1.06, See Appendix A - Activity Preparation Guide 活动 1.06 看附录 A 活动准备指南	This objective has an activity. 此课程目标有一个活动	
1.06.05 Using Water Testing Strips 使用水质检测试纸	A. Recognize that water testing strips are a recommended method for assessing Chlorine, Chlorides, and other parameters of a water sample A. 辨别水质检测试纸为测量水质样本中的氯,氯化物及其他水中参数的建议方法		Question ID: 100010842739 Which of the following testing methods is recommended for use by a coffee equipment technician to measure chlorine levels of a water sample? <u>Test strips</u> Drop-count titration TDS meter pH meter 问题编号: 100010842739	

			下列哪一个为建议咖啡设备技师用来测量水质样本中氯含量的方式? <u>检测试纸</u> 滴定法 TDS 测量仪 pH 仪	
			Question ID: 100010842740 Which two (2) of the following testing methods may be used by a coffee equipment technician to measure the pH of a water sample? <u>Test strips</u> <u>pH meter</u> TDS meter Refractometer 下列的测量方式中,哪二个是咖啡设备技师测量水样本 pH 值的方法? <u>检测试纸</u> <u>pH 仪</u> TDS 测量仪 屈光仪	
			Question ID: 100010842741 Which two (2) of these coffee brewing water parameters are often measured using test strips? <u>Chlorine</u> <u>pH</u> Total Dissolved Solids Temperature 问题编号: 100010842741 下列哪二个咖啡冲煮水的参数通常用检测试纸加以测量? <u>氯</u> <u>pH 值</u> 总溶解固体含量 温度	

	B. Demonstrate the proper use of test strips to measure the chlorine level of a water sample. B.示范正确使用检测试纸来测量水质样本中的氯含量	Activity 1.06, See Appendix A - Activity Preparation Guide 活动 1.06 看附录 A 活动准备指南	This objective has an activity. 此课程目标有一个活动	
1.06.06 Using the SCA Water Chart 使用 SCA 水质图	A. Demonstrate the calculation to convert measurements from units of Degrees of hardness or Grains per Gallon to ppm CaCO₃	Activity 1.06, See Appendix A - Activity Preparation Guide 活动 1.06 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
	B. Demonstrate the proper technique for recording total hardness and alkalinity measurements on the SCA Water Chart A. 示范将硬度的测量值由单位德制硬度,克/美式加仑转换为 PPM 碳酸钙的计算方式 B. 示范在 SCA 水质图上纪录总硬度和碱度测量值的正确技巧	Activity 1.06, See Appendix A - Activity Preparation Guide 活动 1.06 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	

1.07 | Section | Water Treatment System Maintenance 1.07 | 章节 | 水处理设备系统的保养维修
3 Topic 三个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线笔试问题	Resources 相关资源
1.07.01 Water Treatment System Maintenance Requirements 水处理系统维护的需求	A. Recognize that water treatment system media have a fixed capacity and limited life span and must be periodically replaced or recharged A. 辨别水处理系统滤芯有一定的使用量限制及有限的寿命,必需周期性地更新及置换		Question ID: 100010843082 Which of the following factors should be considered when determining if a water treatment system cartridge should be replaced? The amount of time the cartridge has been in service The amount of water which the cartridge has treated The amount of dissolved material in the incoming water <u>All of the above</u> 问题编号: 100010843082 下列哪些因素被认为可决定水处理系统的滤芯需要更换? 滤芯已使用时间 滤芯已处理的水量 进水水质的矿物质溶解量 <u>以上皆是</u> Question ID: 100010843084 What is the likely result of using a water softening cartridge beyond its rated capacity? <u>Reduced softening performance</u> Increased softening performance No change in softening performance Increased water flow rate 问题编号: 100010843084 当水质软化滤芯超过它的使用量限制,则可能会有什么样的结果? <u>较差的软化功效</u>	

			加强的软化功效 在水质软化功效上没有改变 增加水的流过率	
			Question ID: 100010843088 Is the following statement true or false? Water treatment system cartridges have a limited life span and should be replaced as part of a preventive maintenance plan. <u>True</u> False 问题编号: 100010843084 以下论述正确与否?水处理设备滤芯有其有效的使用寿命且应该以更换作为预防性维护计划的一部分 <u>正确</u> 错误	
1.07.02 Maintaining a Quick-change Cartridge System 维护快拆滤芯系统	A. List the common steps for replacing quick-change cartridges, including: 1. Turn off water supply to the system 2. Remove the spent cartridge 3. Install the new cartridge 4. Restore water supply to the system 5. Flush the system as recommended 6. Mark the installed and recommended replacement dates on the cartridge A. 列出一一般置快拆滤芯的步骤,包含: 1. 将系统的水供给截断		Question ID: 100010843090 Which of the following steps is usually recommended before removing a water treatment system cartridge? <u>Turn off water supply to the system</u> Flush water through the system Make sure the water supply is on None of the above 问题编号: 100010843090 在移除水处理设备的滤芯前,下列步骤中哪一个是通常建议的? <u>将系统的水供给截断</u> 以水冲洗整个系统 确认水供给是有的 以上皆非	

	2. 将用过的滤芯拆下 3. 换上新的滤芯 4. 恢复水供给 5. 按建议方式来冲洗系统 6. 将此次滤芯更换日期及下次建议更换的日期纪录于滤芯上		Question ID: 100010843092 Which of the following steps is usually recommended after installing a new carbon filter system cartridge? <u>Flush water through the system</u> Turn off water supply to the system Make sure the water supply is on None of the above 问题编号: 100010843092 在安装上新的活性炭过滤滤芯后,下列步骤中哪一个通常是建议的? <u>以水冲洗整个系统</u> 将系统的水供给截断 确认水供给是有的 以上皆非 Question ID: 100010843093 Is the following statement true or false? All water treatment system heads include a feature which stops water flow through the system when a cartridge is removed. <u>False</u> True 问题编号: 100010843093 以下论述正确与否?所有的水处理系统滤头都会在滤芯移除后自动停止水流经系统中 <u>错误</u> 正确	
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1.07.03 Maintaining a Drop-in Cartridge System 维护置入式过滤滤芯系统	A. List the common steps for replacing drop-in cartridges, including: 1. Turn off water supply to the system 2. Release system pressure 3. Unscrew the housing from the system head 4. Remove the spent cartridge and o-ring 5. Clean and sanitize the housing 6. Install a new cartridge into the housing 7. Lubricate and install an o-ring into the housing 8. Reassemble housing to the system head 9. Restore water supply to the system 10. Flush the system as recommended 11. Check housing seal for leaks 12. Record the installed and change date A. 列出一般更换置入式过滤滤芯的方法,包含: 1. 将系统的水供给截断 2. 释出系统压力 3. 将滤水器外壳自滤头上松开 4. 将用过的滤芯及 O 环移除 5. 清洁并消毒滤水器外壳 6. 将新滤芯装进滤水器外壳 7. 润滑并将 O 环放回滤水器外壳 8. 将滤水器外壳锁回系统滤头 9. 恢复系统水供给 10. 按建议以水冲洗系统 11. 确认滤水器外壳锁紧无泄露		Not tested. 无测验	
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	12. 纪录此次滤芯更换日期及下次预计更换日期			
	B. Demonstrate steps for replacing a drop-in cartridge B. 示范更换置入式滤芯的步骤	Activity 1.07.03, See Appendix A - Activity Preparation Guide 活动 1.07.03 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	

1.08 | Section | Preventive Maintenance Components 1.08 | 章节 | 预防性维护零件
5 Topics 五个主题

Topic 主题	Objectives 目标	AST Notes 讲师	Written Online Exam Questions 在线考试问题	Resources 相关资源
1.08.01 Espresso Machine Brewing Group Components 义式咖啡机冲煮头零件	A. Identify common parts of an espresso machine brewing group A. 确认义式咖啡机冲煮头的一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide	This objective has an activity. 此课程目标有一活动	
	B. List espresso machine brewing group components which are often replaced during a PM service, including: - Group gasket - Group screen - Group screw - Portafilter basket - Portafilter basket retaining spring B. 列出通常由预防性维护服务所更换的义式咖啡机冲煮头零件,包含: -冲煮头垫圈 -冲煮头分水网 -分水网螺丝		Question ID: 100010843102 Which of the following espresso machine brewing group parts are often replaced during a preventive maintenance visit? Group gasket Group screen Group screw <u>All of the above</u> 问题编号: 下方义式咖啡机冲煮头零件中,哪一个通常会在预防性维护服务的时候更换 冲煮头垫圈 冲煮头分水网 分水网螺丝 <u>以上皆是</u>	

	-把手滤篮 -把手滤篮固定弹簧 C. List espresso machine brewing group components which are typically cleaned and inspected during a PM service, including: - Group head - Dispersion plate/disc - Locking bezel - Brew valve - Portafilter C. 列出通常由预防性维护服务提供清洁及检查的义式咖啡机冲煮头零件 -冲煮头 -分水块 -固定框架 -冲煮阀 -冲煮把手		Question ID: 100010843104 Which of the following espresso machine brewing group parts are often cleaned and inspected during a preventive maintenance visit? Group head Dispersion disc Portafilter <u>All of the above</u> 问题编号: 100010843104 下方义式咖啡机冲煮头零件中,哪一个通常会在预防性维护服务的时候清洁检查? 冲煮头 分水块 冲煮把手 <u>以上皆是</u> Question ID: 100010843105 Which of the following espresso machine brewing group parts are usually not replaced during a preventive maintenance visit? <u>Portafilter</u> Group gasket Group screen Portafilter basket 问题编号: 100010843105 下方义式咖啡机冲煮头零件中,哪一个通常不会在预防性维护服务的时候更换 <u>过滤把手</u>	
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			冲煮头垫圈 冲煮头分水网 分水网螺丝 以上皆是	
1.08.02 Espresso Machine Steam Valve Components 义式咖啡机蒸汽阀门零件	A. Identify common parts of an espresso machine steam valve. A. 辨认义式咖啡机蒸汽阀门的一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide 活动 1.10 看附录 A 活动准备指南	This objective has an activity. 这个课程目标有一个活动	
	B. Identify common parts of a ball-end style espresso machine steam wand joint assembly. B 辨认义式咖啡机的球型接头蒸汽棒的一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide 活动 1.10 看附录 A 活动准备指南	This objective has an activity. 这个课程目标有一个活动	
	C. List espresso machine steam valve components which are typically replaced during a PM service, including: - seals - o-rings - gaskets - springs - bushings C. 列出通常由预防性维护服务所更换的义式咖啡机蒸汽阀门零件, 包含: - 止漏胶垫 - O 形环 - 垫片 - 弹簧		Question ID: 100010843106 Which of the following espresso machine steam valve or wand parts are often replaced during a preventive maintenance visit? <u>Steam valve seal</u> Steam valve handle Steam wand Steam wand tip 问题编号: 100010843106 下列义式咖啡机蒸汽阀或蒸汽棒零件中, 哪一个预防性维护服务时需更换的? <u>蒸汽止漏胶垫</u> 蒸汽扳杆 蒸汽棒 蒸汽喷头	

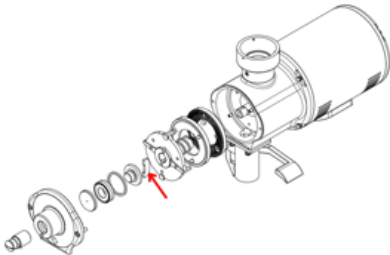
	- 套管 D. List espresso machine steam valve components which are typically cleaned and inspected during a PM service, including: - valve bodies - stems - handles - ball ends D. 列出通常由预防性维护服务所检查及清洁的义式咖啡机蒸气阀门零件,包含: -蒸气阀体 -蒸气阀顶杆 -扳杆 -球头		Question ID: 100010843107 Which of the following espresso machine steam valve or wand parts are often cleaned and inspected during a preventive maintenance visit? <u>Steam valve body</u> Steam valve seal Steam valve o-rings Steam wand o-rings 下列义式咖啡机蒸气阀或蒸气棒零件中,哪一个是预防性维护服务时需进行清洁检查的? <u>蒸气阀体</u> 蒸气止漏胶垫 蒸气阀中的 O 型环 蒸气棒中的 O 型环 Question ID: 100010843108 Which of the following espresso machine steam valve or wand parts are often replaced during a preventive maintenance visit? Steam valve seal Steam valve o-rings Steam wand o-rings <u>All of the above</u> 问题编号: 100010843108 下列义式咖啡机蒸气阀或蒸气棒零件中,哪一个是预防性维护服务时需更换的? 蒸气止漏胶垫 蒸气阀中的 O 型环 蒸气棒中的 O 型环 <u>以上皆是</u>	
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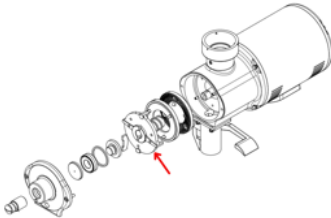
1.08.03 Espresso Machine Steam Boiler Components 义式咖啡机的蒸气锅炉零件	A. Identify common parts of an espresso machine steam boiler which are typically inspected or maintained during a PM service A. 办认通常由预防性维护服务检查或维护的义式咖啡机蒸气锅炉一般零件.	Activity 1.10, See Appendix A - Activity Preparation Guide 活动 1.10 看附录 A 活动准备指南	This objective has an activity. 这个课程目标有一个活动	
	B. List common parts of an espresso machine steam boiler which are typically inspected or maintained during a PM service, including: - Pressure safety valve - Anti-vacuum valve - Water level probe - Pressure switch B. 列出通常由预防性维护服务检查或维护的义式咖啡机蒸气锅炉一般零件, 包含: -压力安全阀 -真空排除阀 -水位探针 -蒸气压力接触开关		Question ID: 100010843110 Which of the following espresso machine steam boiler parts are often inspected during a preventive maintenance visit? Pressure safety valve Anti-vacuum valve Water level probe <u>All of the above</u> 问题编号: 100010843110 下列义式咖啡机锅炉零件中,哪一个预防性维护服务时需检查的? 压力安全阀 真空排除阀 水位探针 <u>以上皆是</u>	
			Question ID: 100010843111 When inspecting an espresso machine anti-vacuum valve, which two (2) of the following signs indicate that the component is working properly? <u>The valve does not leak steam when sealed</u> <u>The valve allows some steam to escape when the stem is depressed</u> The valve allows some steam to escape when sealed	

			The valve remains sealed when the stem is depressed 问题编号: 100010843111 当检查义式咖啡机的真空排除阀时,下面哪二个状态可以判断零件正常运作? <u>当锁紧时阀门并无泄露</u> <u>当蒸气阀顶杆压下时,阀门可让一些蒸气散逸而出</u> 当锁紧时阀门可让一些蒸气散逸而出 当蒸气阀顶杆压下时,阀门仍然锁紧 Question ID: 100010843112 When inspecting an espresso machine pressure safety valve, which two (2) of the following signs indicate that the component should be replaced? <u>The valve allows some steam to escape at normal boiler operating pressure</u> <u>The manufacturer's tamper-resistant feature has been removed or damaged</u> The valve remains sealed at normal boiler operating pressure The valve allows steam to escape at the rated release pressure 问题编号: 100010843112 当检查义式咖啡机的安全阀时,下面哪二个状态可以判断零件需要更换? <u>在正常锅炉运作压力下,阀门可以让一些压力散逸而出</u> <u>制造商在安全阀上的印记标章被移除或破坏</u> 在正常锅炉运作压力下阀门完全紧闭 蒸气从阀门以泄压速度散逸而出	
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1.08.04 Batch Brewer Components 滴滤式咖啡机零件	A. Identify common parts of a batch coffee brewer which are typically inspected or maintained during a PM service.	Activity 1.10, See Appendix A - Activity Preparation Guide 活动 1.10 看附录 A 活动准备指南	This objective has an activity. 这个课程目标有一个活动	
	B. List common parts of a batch coffee brewer which are typically inspected or maintained during a PM service, including: - Water level probe - Spray head A. 办认通常由预防性维护服务检查或维护的滴滤式咖啡机一般零件,包含: -水位探针 -喷洒头		Question ID: 100010843118 Which of the following batch coffee brewer parts are typically inspected and cleaned during a preventive maintenance visit? <u>Water level probe</u> Heating element Brew dispensing valve Solid state relay 问题编号: 100010843118 下列滴滤式咖啡机零件中,哪一个是预防性维护服务时需清洁检查的? <u>水位探针</u> 加热零件 冲煮分配阀 固态继电器 Question ID: 100010843119 Which two (2) of the following batch coffee brewer parts are typically inspected and cleaned during a preventive maintenance visit? <u>Spray head</u> <u>Water level probe</u> Heating element Auto fill valve 问题编号: 100010843119	

			下列滴滤式咖啡机零件中,哪二个是预防性维护服务时需清洁检查的? <u>喷洒头</u> <u>水位探针</u> 加热零件 自动补水阀	
			Question ID: 100010843120 Is the following statement true or false? Batch coffee brewers are simple devices which do not require preventive maintenance. <u>False</u> True 问题编号: 100010843120 以下论述正确与否?批量咖啡冲煮机是简单的装置,无需预防性维护. <u>错误</u> 正确	
1.08.05 Coffee Grinder Components 咖啡磨豆机的零件	A. Identify common parts of coffee grinder which are typically inspected or maintained during a PM service. A. 确认通常由预防性维护服务检查或维护的磨豆机一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide 活动 1.10 看附录 A 活动准备指南	This objective has an activity. 这个课程目标有一个活动	
	B. List common parts of coffee grinder which are typically inspected or maintained during a PM service, including: - Bean hopper - Grinding chamber - Grinding burrs - Burr carrier - Burr carrier shear plate*	*Note that grinder designs vary by manufacturer and not all features are found on all grinders *请注意磨豆机的设计因制造商而异,并非所有的特点都可以在所有的磨豆机上找到	Question ID: 100010843121 Which of the following espresso grinder parts are often inspected and cleaned during a preventive maintenance visit? Grinding burrs De-clumping device Bean hopper <u>All of the above</u>	Ersatzteilliste Spare parts list EK43_EK43S DE_EN_V2.0.3.indd (shopify.com)

	- Outlet chute - Declumping device* B. 列出通常由预防性维护服务检查或维护的磨豆机一般零件,包含 -咖啡豆槽 -研磨腔室 -咖啡刀盘 -刀座 -刀座插销 -出粉通道 -防结块装置		问题编号: 100010843121 下列咖啡磨豆机零件中,哪一个是预防性维护服务时需清洁检查的? 磨豆机刀盘 防结块装置 咖啡豆槽 <u>以上皆是</u> Question ID: 100010843122 Which coffee grinder part is indicated by the arrow in this parts diagram?  <u>Burr carrier shear plate</u> Burr carrier Grinding burr Grinding chamber 问题编号: 100010843122 图中箭头所指是咖啡磨豆机的哪一个部分? <u>刀座插销</u> 刀座 刀盘 研磨腔室	
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			Question ID: 100010843123 Which coffee grinder part is indicated by the arrow in this parts diagram?  <u>Burr carrier</u> Grinding burr Bean hopper Grinding chamber 问题编号: 100010843123 图中箭头所指是咖啡磨豆机的哪一个部分? <u>刀座</u> 刀盘 咖啡豆槽 咖啡粉槽	
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1.09 | Section | Maintenance Strategies 1.09 | 章节 | 保养策略
2 Topics 二个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线考试问题	Resources 相关资源
1.09.01 Preventive versus Reactive Maintenance 预防式与回应式的维护	A. Define preventive maintenance as regular and routine service performed to maintain proper system function and reduce unexpected equipment failure. B. Define reactive maintenance as unplanned repair of equipment that has failed in order to restore proper system function. A. 定义预防式维护为定期常规的服务执行以维持恰当的系统功能并减少无法预期的设备故障 B. 定义响应式维护为设备故障后,为了让系统回到正常功能运作而进行未在计划内的修复	Though a good preventive maintenance program should reduce the likelihood of unexpected equipment failure, no maintenance program can completely eliminate this possibility. Most programs seek to balance program cost vs risk of downtime, a topic which will be explored in higher-level courses. 因此,一个良好的预防性维护计划应该会减少无法预期的设备故障发生的可能性,没有任何的维护计划可以完全的消除未预期故障发生的可能性. 多数的计划寻求花费成本与停机时间风险的平衡,这个课题我们会在更高阶课程中再进一步的探讨	Question ID: 100010843124 Which of the following best describes the purpose of coffee equipment preventive maintenance? <u>Regular and routine service to maintain proper system function</u> Unplanned repair of equipment that has already failed Regular service in order to eliminate all possibility of unexpected equipment failure Unnecessary service scheduled in order to increase technician revenue 问题编号: 100010843124 下列何者是咖啡设备预防性维护目的的最佳描述? <u>定期常规的服务执行以维持恰当的系统功能</u> 未在计划内的修理设备的故障 为了减少所有未预期的设备故障而执行的定期服务 为了增加技师的收入而计划不需要的服务	
			Question ID: 100010843125 Which of the following best describes coffee equipment reactive maintenance? <u>Unplanned repair of equipment that has failed in order to restore proper system function</u> Regular and routine service to maintain proper system function Maintenance recommended by the equipment manufacturer	

			according to expected component lifespan Service to replace components which show signs of manufacturing defects 问题编号: 100010843125 下列何者是咖啡设备回应性维护的最佳描述? <u>设备故障后,为了让系统回到正常功能运作而进行未在计划内的修复</u> 定期常规的服务执行以维持恰当的系统功能 设备制造商因应零件的寿命期限而建议的维护 提供服务以更换有制造瑕疵征兆的零件	
			Question ID: 100010843126 Which of the following best describes an espresso machine service which is planned to be performed every six months to rebuild brewing groups and steam valves? <u>Preventive maintenance</u> Reactive maintenance Installation service Warranty service 问题编号: 100010843126 下列何者绝佳的形容一家义式咖啡机设备商计划每六个月提供对冲煮头及蒸气阀门的维护? <u>预防性维护</u> 回应性维护 意大利式维护 保修服务	

1.09.02 Recommended Maintenance Intervals 建议的保养周期	A. Recognize that different components will have different recommended maintenance intervals and expected service lives A. 认知到不同的零件有不同建议的保养周期及可预期的使用寿命 B. List factors which impact the longevity of components,including: - wear due to use and abuse - degradation due to high-temperature operating environment - degradation due to duty cycling or extended operation - loss of function due to scale buildup B. 列出会影响零件寿命的因素 -因使用及滥用而引起的耗损 -因高温的作业环境而引起的损害 -因工作周期或延长的作业而引起的损害 -因水垢而引起的功能丧失 C. Recognize that equipment manufacturers may recommend preventive maintenance schedules for their equipment C. 认知设备制造商可能会建议设备		Question ID: 100010843127 Which of the following factors may affect the expected life span of coffee equipment components? Wear due to normal use High temperature operating environment Buildup of limescale All of the above 问题编号: 100010843127 下列哪一个因素可能会影响咖啡设备零件的寿命期限? 一般使用所引起的损耗 高温作业环境 水垢的累积 以上皆是 Question ID: 100010843128 Is the following statement true or false? Preventive maintenance is not necessary for an espresso machine which is powered on all day, every day, but not used very often. False True 问题编号: 100010843128 以下的论述正确与否?每天且一整天都打开却不常用的义式咖啡机是不需要预防性维护的. 错误 正确	
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	所需的预防性维护周期		Question ID: 100010843129 Which two (2) statements best describe recommended preventive maintenance for two identical espresso grinders, one which grinds 10kg of coffee per week and the other which grinds 2kg per week? <u>Regular preventive maintenance for both grinders</u> <u>Different grinder burr replacement schedules for each grinder</u> Regular preventive maintenance for one grinder only The same grinder burr replacement schedule for both grinder 问题编号: 100010843129 二台相同义式磨豆机,一台每周磨 10kg 咖啡,另一台每周磨 2kg 咖啡,下列哪二个论述最好的描述了它们的预防性维护? <u>二台磨豆机都需要进行周期性的预防性维护</u> <u>每台磨豆机有不同的刀盘更换周期</u> 只有一台磨豆机需要周期性的预防性维护 二台磨豆机都适用一样的刀盘更换周期	
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1.10 | Section | Preventive Maintenance Tools and Supplies 1.10 | 章节 | 预防性维护的工具与零件消耗品

2 Topics 二个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线笔试问题	Resources 相关资源
1.10.01 Recommended Hand Tools 建议的手动工具	A. List common hand tools used for maintaining espresso machine brewing group, including: - Pick awl - Short screwdriver - Stiff bristle cleaning brush - Blank portafilter basket A. 列出一般维护义式咖啡机冲煮头时所需手动工具, 包含: - 尖锥 - 短柄螺丝起子 - 硬毛清洁刷 - 无孔滤杯 B. List common hand tools used for maintaining espresso machine steam valve, including: - Wrenches - Pick awl - Stiff bristle cleaning brush - Snap ring pliers B. 列出一般维护义式咖啡机蒸气阀时所需手动工具, 包含: - 扳手 - 尖锥 - 硬毛刷 - 卡环钳 C. List common hand tools used for		Question ID: 100010844140 Which two (2) of the following tools are often recommended for replacing an espresso machine group gasket? <u>Pick awl</u> <u>Stiff-bristle cleaning brush</u> Wrench Snap ring pliers 问题编号: 100010844140 下列哪二个工具通常建议在更换义式咖啡机冲煮头垫圈时使用? <u>尖锥</u> <u>硬毛刷</u> 扳手 卡环钳	
			Question ID: 100010844141 Which of the following tools is often recommended for replacing coffee grinder burrs? <u>Screwdriver</u> Pick awl Snap ring pliers Wrench 问题编号: 100010844141 下列哪个工具通常建议在更换义式咖啡磨豆机磨盘时使用? <u>螺丝起子</u>	

	maintaining batch brewers, including: - Screwdrivers - Wrenches - Stiff bristle cleaning brush - Large water pitcher or container C. 列出一般维护滴滤式咖啡机时所需手动工具, 包含: -螺丝起子 -扳手 - 硬毛刷 - 大水瓶或水桶 D. List common hand tools used for maintaining a coffee grinder, including: - Screwdrivers - Soft bristle brush - Stiff bristle cleaning brush - Vacuum cleaner D. 列出一般维护义式咖啡磨豆机时所需手动工具, 包含: - 螺丝起子 - 软毛刷 - 硬毛刷 - 吸尘器		尖锥 卡环钳 扳手 Question ID: 100010844142 Which of the following tools is often recommended for servicing a ball-joint style steam wand? <u>Wrench</u> Pick awl Screwdriver Snap ring pliers 问题编号: 100010844142 下列哪个工具通常建议在更换义式咖啡机球型接头蒸气棒时使用? <u>扳手</u> 尖锥 螺丝起子 卡环钳	
1.10.02 Recommended Consumables 建议消耗品	A. List the recommended supplies used to clean an espresso machine and batch brewer, including: - Espresso machine detergent - Coffee brewer detergent - Milk system cleaning solution	Examples of o-ring lubricant include Dow 111/Molykote and Examples of grinder collar thread grease include Kluberlub NH1 11/222 举例 O 环润滑油包含道康宁 Dow	Question ID: 100010844197 Which of the following cleaning supplies is recommended when servicing an espresso machine brewing group? <u>Espresso machine detergent</u> Milk system cleaner Kitchen equipment degreaser	

	B. List the types of lubricants which may be recommended for maintaining espresso machines and coffee grinders, including: - Silicone o-ring lubricant - Grinder collar thread grease* A. 列出一般建议用于清洁一台义式咖啡机及滴滤式咖啡机所需消耗品, 包含: - 义式咖啡清洁剂 - 滴滤式咖啡机清洁剂 - 牛奶系统的清洁溶剂 B. 列出可能适用于维护义式咖啡机及义式咖啡磨豆机的润滑剂类型, 包含: - 硅 O 环润滑油 - 上刀座固定螺纹润滑油	111/Molykote, 上刀座固定螺纹润滑油包含克鲁勃 Klüberlub NH1 11/222。	WD-40 问题编号: 100010844197 下列哪一个清洁消耗品建议用于维护义式咖啡机冲煮头? <u>义式咖啡机清洁剂</u> 牛奶系统清洁剂 厨房设备除油剂 WD-40 防锈润滑剂 Question ID: 100010844247 Which of the following characteristics should lubricants used with espresso machine o-rings have? Suitable for high-temperature applications Approved for incidental food contact Suitable for contact with water <u>All of the above</u> 问题编号: 100010844247 下列哪一个特性是用于义式咖啡机 O 型环润滑油应具备的? 适用于高温的应用 核准可偶发性的与食物接触 适于与水接触 <u>以上皆是</u> Question ID: 100010844248 Is the following statement true or false? Lubricant used on a threaded espresso grinder adjustment collar should be approved for incidental food contact. <u>True</u> False	
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			问题编号: 100010844248 下列论述正确与否? 使用于义式磨豆机上刀座固定螺纹的润滑剂需核准可偶发性的与食物接触 <u>正确</u> 错误	
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1.11 | Section | Performing Maintenance1.11 | 章节 | 维护的实施
7 Topics 七个主题

Topic 主题	Objectives 目标	AST Notes 讲师附注	Written Online Exam Questions 在线笔试问题	Resources 相关资源
1.11.01 Safety During a Preventive Maintenance Service 预防性维护服务过程的安全	A. State that, to minimize potential injury, coffee equipment must be powered off and unplugged whenever maintenance requires that panels be removed, or grinders be disassembled A. 指出为了降低潜在伤害, 不论维护需要拿掉隔板或拆开磨豆机, 咖啡设备必需在关闭电源且拔掉插头下进行		Question ID: 100010844257 Which of the following espresso machine preventive maintenance activities require the equipment to be powered off and unplugged? <u>Inspecting pressure switch contacts</u> Cleaning an espresso machine group Replacing espresso machine group gaskets Adjusting pump pressure 问题编号: 100010844257 下列哪一个义式咖啡机的预防性维护活动需要将设备关闭电源并拔掉插头? <u>检查压力接触开关</u> 清洁义式咖啡机冲煮头 更换义式咖啡机冲煮头垫圈 调整泵浦压力	

			Question ID: 100010844259 Which of the following batch brewer preventive maintenance activities require the equipment to be powered off and unplugged? <u>Removing and cleaning level probe</u> Inspecting hot water tap function Removing and cleaning spray head Confirming brew volumes 问题编号: 100010844259 下列哪一个滴滤式咖啡机的预防性维护活动需要将设备关闭电源并拔掉插头? <u>移除并清洁水位探针</u> 检查热水出水功能 移除并清洁喷洒头 确认冲煮水量	
			Question ID: 100010844260 Which of the following coffee grinder preventive maintenance activities require the equipment to be powered off and unplugged? <u>Replacing grinder burrs</u> Adjusting grinder setting Calibrating grind adjustment knob Adjusting programmed grinding time 问题编号: 100010844260 下列哪一个咖啡磨豆机的预防性维护活动需要将设备关闭电源并拔掉插头? <u>更换研磨刀盘</u> 调整研磨设定	

			校正研磨刻度调整旋钮 调整计划性的研磨时间	
1.11.02 Preventive Maintenance Checklist 预防性维护检查表	A. Recognize that a preventive maintenance checklist is recommended to ensure that no maintenance steps are missed and provide a record of services performed A. 认知预防性维护检查表建议用来确保没有任可一个维护步骤会被跳过且提供服务实施的纪录		Not tested. 无测验	
	B. Demonstrate the use of a preventive maintenance checklist to record preventive maintenance service activities. B. 示范预防性维护检查表的使用,以纪录预防性维护服务活动	Activity 1.13.02, See Appendix A - Activity Preparation Guide 活动 1.13.02 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
1.11.03 Espresso Machine Preventive Maintenance Cleaning Steps 义式咖啡机预防性维护清洁步骤	A. List common cleaning steps which are recommended during an espresso machine preventive maintenance service, including: - Cleaning the brewing group and dispersion disc - Cleaning portafilter - Detergent backflush the brewing group - Cleaning steam wand and tip A. 列出义式咖啡机预防性维护服务时,一般建议进行的清洁步骤,包含: -清洁冲煮头及分水块		Question ID: 100010844421 Which of the following cleaning activities are recommended as part of an espresso machine preventive maintenance service? Cleaning the brewing group Detergent backflush the brewing group Cleaning steam tip <u>All of the above</u> 问题编号: 100010844421 下列何者的清洁活动建议为义式咖啡机预防性维护服务的一部分?	

	-清洁冲煮把手 -以清洁剂逆洗冲煮头 -清洁蒸气棒及喷头		清洁冲煮头 以清洁剂逆洗冲煮头 清洁蒸气棒及喷头 <u>以上皆是</u> Question ID: 100010844429 Which of the following cleaning activities are NOT usually recommended as part of an espresso machine preventive maintenance service? <u>Disassemble and clean brew valve</u> Cleaning the brewing group Cleaning the portafilter Detergent backflush the brewing group 问题编号: 100010844429 下列何者的清洁活动通常不建议为义式咖啡机预防性维护服务的一部分? <u>拆开并清洁冲煮阀</u> 清洁冲煮头 清洁冲煮把手 以清洁剂逆洗冲煮头 Question ID: 100010844433 From the drop-down menu, choose the correct step to fill in the blank to reflect recommended espresso machine brewing group backflushing procedure: 1. Insert blank basket or cleaning disk into portafilter 2. _ 3. Insert portafilter into group	
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			4. _ 5. Remove portafilter and rinse group and portafilter 6. _ 7. Activate brew cycle for 10 seconds, stop 10 seconds, repeat 5 times 8. Replace standard basket in portafilter or remove backflush disk A = Add recommended amount of espresso detergent B = Activate brew cycle for 10 seconds, stop 10 seconds, repeat 5 times C = Insert portafilter without detergent into group <u>A = 2 / B = 4 / C = 6</u> 问题编号: 100010844433 由下拉式选单中,选取正确的步骤填入空白,以反应正确的义式咖啡机冲煮头逆洗流程 1. 将无孔滤杯或逆洗片置入冲煮把手中 2. _____ 3. 将冲煮把手卡入冲煮头 4. _____ 5. 移开把手并冲洗冲煮头及冲煮把手 6. _____ 7. 启动 10 秒的冲煮,然后停止 10 秒,这样的循环持续五次 8. 将冲煮把手中的滤杯更换回原来的标准滤杯或移除逆洗片 A=加入建议适量的义式咖啡清洁剂 B=启动 10 秒的冲煮,然后停止 10 秒,这样的循环持续五次 C=将没有清洁剂的冲煮把手卡入冲煮头 A = Add recommended amount of espresso detergent	
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			B = Activate brew cycle for 10 seconds, stop 10 seconds, repeat 5 times C = Insert portafilter without detergent into group	
	B. List the steps to backflush an espresso machine brewing group B. 列出义式咖啡机冲煮头的逆洗步骤	Activity 1.13.03, See Appendix A - Activity Preparation Guide 活动 1.13.03 看附录 A 活动准备指南	This objective has an activity. 这个课程目标有一个活动	
1.11.04 Espresso Machine Preventive Maintenance Parts Replacement	A. Demonstrate espresso machine group screen and gasket replacement	Activity 1.13.04-1.13.05, See Appendix A - Activity Preparation Guide 活动 1.13.04-1.13.05 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
义式咖啡机预防性维护-零件更换	B. Demonstrate espresso machine steam wand joint rebuild A. 示范义式咖啡机冲煮头的分水网及垫圈的更换方式 B. 示范义式咖啡机蒸气棒接头的重新组装	Activity 1.13.04-1.13.05, See Appendix A - Activity Preparation Guide 活动 1.13.04-1.13.05 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
1.11.05 Espresso Machine Preventive Maintenance Inspection Steps 义式咖啡机预防性维护检查步骤	A. List common inspection steps which are recommended during an espresso machine preventive maintenance service, including: - Overall steam leaks - Safety valve steam leaks - Steam valve leaks - Overall water leaks - Brew valve leaks - Anti-vacuum valve function - Pressure switch contact condition - Drain system function	If inspection shows that these operating conditions are not within desired range, adjustment may be required to correct them. Adjustment is covered in higher-level course 如果检查显示这些运作的状况不在预期的范围内,可能会需会调整来加以修正 调整的内容将在更高级数的课程中	Question ID: 100010844437 Which of the following espresso machine components should be inspected for possible steam leaks during a preventive maintenance service? Safety valve Steam valve Anti-vacuum valve <u>All of the above</u> 问题编号: 在预防性维护服务中,为了蒸气泄露的可能性,需要检查下	

	- Cold water expansion valve function A. 列出一般义式咖啡机预防性维护服务所建议的检查步骤,包含: - 所有蒸气的泄出 - 安全阀门蒸气的泄出 - 蒸气阀的泄出 - 所有水的排出 - 冲煮阀的排水 - 真空排除阀的功能 - 压力接触开关的状况 - 排水系统功能 -冷水膨胀阀的功能 B. List common espresso machine operating conditions which should be recorded during a preventive maintenance service, including: - Pump pressure - Steam boiler pressure - Group water temperature - Brew volumes (auto-volumetric machines) - Group water debit B. 列出在预防性维护服务中,应该被纪录下的一般义式咖啡机操作状况,包含: - 泵浦压力 - 蒸气锅炉压力 - 冲煮头的水压 - 冲煮水量(自动定量咖啡机)		列哪一个义式咖啡机零件? 安全阀 蒸气阀 真空排除阀 <u>以上皆是</u> Question ID: 100010844439 When inspecting the espresso machine for water leaks, for which of the following components is dripping a small amount of water acceptable? <u>Cold water expansion valve</u> Brew valve Steam valve Steam joint 问题编号: 100010844439 当检查义式咖啡机的排水时,哪一个零件滴出小小量的水是可以接受的? <u>冷水膨胀阀</u> 冲煮阀 蒸气阀 蒸气棒接头 Question ID: 100010844441 Which of the following espresso machine operating conditions should be inspected during preventive maintenance service? Pump pressure Steam boiler pressure Group water debit <u>All of the above</u>	
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			问题编号: 100010844441 下列哪一个义式咖啡机的运作状况在预防性维护服务时需要检查? 泵浦压力 蒸气锅炉压力 冲煮头的出水量 <u>以上皆是</u>	
	C. Demonstrate the inspection of an espresso machine during preventive maintenance service C. 示范预防性维护服务中所进行的义式咖啡机检查	Activity 1.13.04-1.13.05, See Appendix A - Activity Preparation Guide 活动 1.13.04-1.13.05 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
1.11.06 Batch Brew Preventive Maintenance 滴滤式咖啡机的预防性维护	A. List common maintenance steps which are recommended during a batch brewer preventive maintenance service, including: - Inspect and clean water level probe - Inspect and clean spray head - Inspect hot water tap function - Confirm brew volumes A. 列出一一般滴滤式咖啡机进行预防性维护服务的建议维护步骤, 包含 -检查并清洁水位探针 -检查并清洁喷洒头 -检查热水出水功能 -确认冲煮水量		Question ID: 100010844445 Which of the following maintenance steps are NOT generally recommended during a batch brewer preventive maintenance service? <u>Drain and refill the water tank</u> Inspect and clean water level probe Inspect and clean spray head Confirm brew volumes 问题编号: 100010844445 在滴滤式咖啡机的预防性维护服务中,下列哪一个维护步骤通常并不建议? <u>将热水槽排干然后重新装水</u> 检查并清洁水位探针 检查并清洁喷洒头 确认冲煮水量	

			Question ID: 100010844447 Which of the following maintenance steps is recommended for a coffee batch brewer water level probe during preventive maintenance service? <u>Inspect and clean to remove buildup</u> Trim at least 1 cm from the end Replace it Check its capacitance with a multimeter 问题编号: 100010844447 在滴滤式咖啡机的预防性维护服务中,下列哪一个维护步骤是建议对水位探针进行的? 检查并清洁以去除水垢 从尾段修剪 1cm 更换它 用万用表检查它的电容	
			Question ID: 100010844449 Which of the following maintenance steps is NOT recommended for a coffee batch brewer spray head during preventive maintenance service? <u>Use a caliper to measure all holes against factory specifications</u> Inspect and replace if damaged Clean to remove coffee residue Remove any limescale buildup 问题编号: 100010844449 在滴滤式咖啡机的预防性维护服务中,下列哪一个维护步骤不建议对喷洒头进行?	

			<u>使用测径仪来测量所有的洞与工厂规格作参照</u> 如果毁损需进行检查且更换 清洁并去除所有的咖啡残留物 去除所有的水垢	
	B. Demonstrate inspection and cleaning of a batch brewer level probe B. 示范检查并清洁滴滤式咖啡机的水位探针	Activity 1.13.06, See Appendix A - Activity Preparation Guide 活动 1.13.06 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	
1.11.07 Espresso Grinder Preventive Maintenance 义式磨豆机的预防性维护	A. List common maintenance steps which are recommended during espresso grinder preventive maintenance service, including: - Clean bean hopper - Clean grind chamber throat - Clean and inspect grinder burrs - Replace grinder burrs as-needed - Clean inside grinding chamber - Clean and re-lubricate adjustment collar threads - Calibrate and adjust grind setting per customer specification - Adjust grinding time per customer specification (on-demand grinders only) A. 列出一般义式咖啡磨豆机的预防性维护服务建议的维护步骤, 包含: -清洁咖啡豆槽 -清洁研磨腔室颈 -清洁并检查研磨刀盘		Not tested. 无测验	

	-如果需要更换刀盘 -清洁研磨腔室的内部 -上刀座固定螺纹的清洁并重新上润滑油 -校正并根据客户的需求调整研磨设定 -根据客户的需求调整研磨时间(只有定量磨豆机)			
	B. Demonstrate replacement of espresso grinder burrs. B. 示范更换义式咖啡机刀盘	Activity 1.13.07, See Appendix A - Activity Preparation Guide 活动 1.13.07 看附录 A 活动准备指南	Tested on Practical Exam. 于实务操作考试中进行测验	http://web.archive.org/web/20210620214141/https://www.espressoparts.com/pages/changing-grinder-burrs-when-and-how

5. Essential SCA Training Documents 必要的 SCA 训练文件

- The SCA Water Chart
- SCA 水质图

6. Bibliography 参考书目

- Wellinger, Marco, Samo Smrke, Chahan Yeretzian, *The SCA Water Quality Handbook*, Specialty Coffee Association, 2018
- Billemon, Ronny, *Equilibrium in Water and Coffee*, 2018
- Colonna-Dashwood, Maxwell, Christopher H. Hendon, *Water for Coffee*, Maxwell Colonna-Dashwood and Christopher H. Hendon, 2015

7. Required Equipment and Supplies List 所需设备及用品清单

The chart below represents the minimum pieces of equipment and supplies for the instructional part of this course. This does not include instructional supplies like clipboards, aprons, and similar. Please refer to the Water & preventive Maintenance Foundation Practical Exam Trainer Guide for additional items required to conduct the practical exam.

IMPORTANT: Be sure to have an easily accessible, fully-equipped first aid kit on site, along with the appropriate personal protection equipment necessary for the activities of the course and exam.

底下的表格代表此课程在教学部分所需设备及零件的最小量. 这并不包含教学需要的板夹,围裙及类似的项目. 其他进行考试所需要的额外项目,请参考水&预防性维护基础实务测验讲师指南,

重要:务必确保现场备有完整配备且方便取用的急救箱,还有课程活动及考试中需要的个人适当的防护配备

Item 品项	Qty 数量	Notes 备注
Coffee Equipment 咖啡设备		
Espresso machine 义式咖啡机	1	Espresso machine must be connected to water supply and electricity 义式咖啡机必需接上水源及电源
Batch Coffee Brewer 滴滤式咖啡机	1	Batch brewer need not be connected to water supply and electricity 滴滤式咖啡机必需接上水源及电源
Espresso Grinder 义式咖啡磨豆机	1	Espresso grinder must be able to be connected to electricity 义式咖啡磨豆机必需接上水源及电源
Espresso Machine Components 义式咖啡机零件		Unless noted, all component quantities are one per (3) three students. In each case, there should be a used/damaged and new component that fits the demonstration espresso machine. 除非另有说明,所有零件数量是三个学生一组. 在每一个项目中,都需要准备与示范义式咖啡机相称的使用过的/损坏的及全新的零件
Group gaskets 冲煮头垫圈	1	Have an example of a previously-used and hardened group gasket as well as new gaskets 准备之前使用的垫圈, 已变硬的垫圈及新的垫圈
Group screen	1	Have an example of a previously-used screen which is deformed or has coffee build-up, as well as new screens

冲煮头分水网		准备之前使用过已变形,使用过仍良好的分水网,及新的分水网
Group screw 分水网螺丝	1	Have an example of a previously-used and damaged screw as well as new 准备之前使用的螺丝,损坏的螺丝及新的螺丝
Portafilter basket 冲煮把手滤杯	1	Have an example of a previously-used and damaged basket as well as new baskets 需准备之前使用的滤杯, 损坏的滤杯及全新滤杯
Portafilter basket retaining spring 冲煮把滤杯固定弹簧	1	Spring must fit portafilter that will be maintained during class activities 弹簧必需与课程活动中维护的把手合适相称
Steam valve rebuild kit 蒸气阀重置组	1	Kit should include common replacement parts to rebuild the steam valve used in class activities 组合需包含一般替换零件以便在课程活动中重新组装蒸气阀使用
Steam wand joint rebuild kit 蒸气棒接头重置组	1	Kit should include common replacement parts to rebuild the steam wand joint used in class activities 组合需包含一般替换零件以便在课程活动中重新组装蒸气棒接头使用
Pressure switch 压力开关	1	One per class, 30A standard style, may be new or used 一个班一个,30A 标准式,可以是全新或使用过的
Espresso Grinder Components 义式磨豆机零件		Unless noted, all component quantities are one per three (3) students. 除非另有说明,所有零件的数量都是三个学生一组
Grinding burrs with screws 磨豆机刀盘带螺丝	1	Display one example each of burrs in good condition and heavily worn burrs, both must be the replacement part for the espresso grinder to be maintained during course activities. 展示样品需包含一个良好的刀盘, 及一个已严重磨损的刀盘。在课程活动中, 二者都必须是待维护义式磨豆机的替换零件。
Hand and Measurement Tools 手动及量测工具		Unless noted, all component quantities are listed per three (3) students. 除非另有说明,所有零件的数量都是三个学生一组
Straight screwdrivers 一字螺丝起子	1	Provide assorted sizes, including short bladed 提供各种尺寸,包含短柄的
Phillips head screwdriver 十字螺丝起子	1	Provide assorted sizes 提供各种尺寸
Pick awl 尖锥	1	Provide assorted styles, including straight and angled 提供各种尺寸,包含直的和有角度的
Stiff-bristled cleaning brush 硬毛刷	1	
Blank portafilter basket	1	

无孔滤杯		
Adjustable wrench 活动扳手	1	
Set of Combination Wrenches 一组的多用扳手	1	Metric or standard in a range that fits equipment used in the course 课程中所使用的为标准或公制扳手，并在一个范围内与设备相符合
Snap ring pliers 卡环钳	1	Either one reversible-style or one each of the internal and external style may be used 可为一支反转式卡环钳或内部及外部卡环钳各一
Soft bristle brush 软毛刷	1	
Small vacuum cleaner 小型吸尘器	1	
Electrical conductivity/TDS meter 导电度计/TDS 测量仪	1	Handheld version 手持式版本
Handheld pH meter 手持式 pH 测量仪	1	
Titration measurement kit for total hardness/alkalinity 总硬度/碱度滴定测量组		This is a drop count kit. Note that this kit is often labeled as measuring carbonate hardness/KH. 这是一个滴定组. 注意滴定组通常标记为测量 碳酸盐硬度/KH
Consumable Materials 消耗性材料		Provide sufficient quantities of consumables for all activities and Practical Exam sections 提供足够的消耗量给所有的活动和实务考试
Espresso machine detergent 义式咖啡机清洁剂		
Coffee brewer detergent 滴滤式咖啡机清洁剂		
Milk system cleaning solution 牛奶系统清洁溶剂		
Silicone o-ring lubricant (NSF)		

硅 O 环润滑油(NSF)		
Grinder collar thread grease (NSF) 上刀座固定螺纹润滑油(NSF)		
Water test strips for Chloride testing 氯测试的水测验试纸		
Personal Protective Equipment 个人防护设备		Unless noted, all component quantities are one per student. 除非另有说明,所有的零件数量皆为一个学生一组
Insulating work gloves 绝缘工作手套	1	Make various sizes available, encourage students to bring their own 提供多种尺寸选择,鼓励学生带他们自己的
Safety glasses 安全眼镜	1	Make various sizes available, encourage students to bring their own 提供多种尺寸选择,鼓励学生带他们自己的

8. Appendices 附录

Appendix A: Activity Preparation Guide 附录 A: 活动准备指南

IMPORTANT SAFETY NOTICE: 重要安全注意事项

This course includes activities which involve the learner interacting with live electricity and hand tools. It is the sole responsibility of the AST to ensure that:

本课程包含的活动让学员与现场电力设备及手动工具有所互动, 讲师最主要的责任即是确保以下注意事项:

- All Learners are properly supervised at all times while working with live electricity and hand tools
- 所有的学员在都在适当的监督下操作电力设备及手动工具
- All Learners use appropriate Personal Protective Equipment during course activities
- 所有学员在课程活动期间皆使用适当的个人防护配备

- All working stations are set up to meet the safe workplace recommendations detailed in guidebook sections 1.01 and 1.06 and all local workplace safety standards
- 所有工作站的设置皆须达到指引手册章节 **1.01**、**1.06** 所载明之安全建议及所有当地工作场域安全标准
- All Learners are introduced to the safe use of all hand and measurement tools to be used prior to beginning
- 在课程开始前,向所有学员介绍如何安全使用所有手动及测量工具
- If unsafe behavior is observed at any time:
- 无论何时,学员一旦违反安全规定,
- the activity is stopped for the Learner
- 应立即中止该学员的活动,
- the Learner is provided with feedback
- 并给予学员反馈,
- the Learner demonstrates an understanding of safe behavior before continuing
- 请学员确实理解何谓安全行为后始可继续参与活动
- All course activities are conducted in accordance with local safety regulations and best practices
- 所有课程活动皆按照当地安全规定及最佳实务进行。
- All stations which involve live electricity are connected in accordance with local electrical code and workplace safety regulations
- 所有需用电之工作站皆须依照当地电工标准及工作场域安全规定进行连接。
- All stations which involve live electricity are connected to power using GFCI devices which have been tested for proper function before beginning this exam.
- 所有需用电之工作站连接到电源皆须使用漏电断路设备,并在测验前通过功能测试。
- All outlets and devices are properly grounded and tested by the AST to confirm proper grounding before beginning this course
- 所有插座及设备在课程开始前皆确实接地,并经由讲师测试及确认。
- All hand tools and measurement devices are inspected for safe and proper function before beginning this course
- 所有手动工具及测量设备在课程开始前皆经测试后确认安全性及功能正常。

If local workplace safety regulations do not allow for a prescribed activity, the AST is not to conduct this activity and is to contact SCA Education to discuss alternative activities

假如当地安全电工标准或工作场域安全规定不允许进行指定的活动,讲师将不进行该活动并连络 **SCA** 教育团队讨论替代方案

Preventive Maintenance Checklist

预防性维护检核表

The Activities and Practical Exam in this course include tasks which are commonly performed during a preventive Maintenance visit. To receive full credit on the Practical Exam the Learner must record all observations, calculations, and task completions on the preventive Maintenance Checklist found on the last page of the Practical Exam Learner Worksheet.

In order to ensure the Learners are familiar with this checklist prior to the Practical Exam, the Trainer must provide this checklist during the Activities below. The Trainer may additionally choose to create a more complete sample checklist for the Learner's future reference, however this checklist should not be used during these Activities or the Practical Exam.

课程中的活动及实务测验包含了预防性维护服务中一般会执行的任务。为了在实务测验中得到满分，学员需在实务测验学员工作表的最后一页-预防性维护检核表上纪录所有的观察、计算、及完成任务。

为了确保学员在实务测验前熟悉预防性维护检核表，讲师必需在下列活动时提供检核表。

讲师可能会额外选择更完整的检核表以利学员将来进一步的参考，然而这份检核表不应在活动或实务测验中使用。

Activity Code 活动代码	Preparation Guide 准备指南
1.02 Water Treatment System Components 水处理系统零件	In this activity, learners interact with and learn to identify physical components from sanitary quick-change cartridge and drop-in cartridge style water treatment systems. Prepare common styles and connect supply lines, being sure to include examples of pressure gages and shut off valves of types commonly used regionally. Systems and components should be dry to enable easy assembly and disassembly. Ensure that the students interact with the drop-in cartridge style system that will be used in Activity 1.07.03 and the Practical Exam. This can be done throughout the course or at one specific activity. Estimated activity time: 15 minutes 在这个活动中，学员将会学习辨认干净的快拆滤芯及置入式水处理系统的实体零件并与之互动。准备常见的型式并接上水源，确认样本中包含区域常见的压力表及关闭阀。系统及零件应该是干燥的，以便可以容易的组装及拆开。确认学员与活动 1.07.3 及实务测验中的置入式滤芯系统有所互动，这可以透过课程或一个指定的活动来达成。 预估活动时间:15 分钟
1.06 Measuring Water Composition 测量水的组成	In this activity, learners practice using several methods to assess the composition of water samples. Trainer must explain, demonstrate, and guide learners through the following tasks: ● Measure Electrical Conductivity and estimated TDS of water using a handheld meter and record on a PM Checklist ● Measure total hardness and alkalinity of water using a drop-count titration test kit and record on a PM Checklist ● Measure pH of water using a handheld meter ● Measure Chlorine level of water using a test strip ● Perform calculation to convert total hardness and alkalinity results from Degrees of hardness or grains per gallon into ppm CaCO₃. ● Record total hardness and alkalinity measurements on the SCA Water Chart and assess whether results are within SCA Core Zone recommendation. ● Trainer must create and provide a visual reference for using the drop-count titration test kit which shows color change expected for each test and conversion factor for ppm CaCO₃ calculation. This visual aid is to be made available during this activity and the Practical Exam. Estimated activity time: 45 minutes 在这个活动中，学员练习使用几种方式以取得水样本的组成。讲师必需透过以下任务解释、示范及教导学员：

	• 使用手持式仪器测量导电度、估测水样本中的 TDS 并纪录在预防性维护检核表上 • 使用滴定测试组合测量总硬度及碱度并纪录在预防性维护检核表上 • 使用手持式仪器测量水样本的 pH 值 • 使用检测试纸来测量水样本的氯含量 • 执行计算以便将总硬度及碱度由测量结果的德制硬度或克/加仑转换成 ppm CaCO₃. • 不论测量结果是否在 SCA 水质核心区的建议内，在 SCA 水质图上纪录总硬度及碱度的测量结果 • 讲师必需提供标示出各个滴定试验所预计的颜色变化之视觉参照和 ppm CaCO₃ 的换算系数。此视觉参照供学生于本活动及实务测验中使用。 预估活动时间:45 分钟
1.07.03 Maintaining a Drop-in Cartridge System 维护置入式滤芯系统	In this activity, learners practice replacing the cartridge in a drop-in style water treatment system. Trainer must explain, demonstrate, and guide learners through the following tasks: • Turn off water supply to the system • Release system pressure • Unscrew the housing from the system head • Remove the spent cartridge and o-ring (if used) • Clean and sanitize the housing • Install a new cartridge into the housing • Lubricate and install an o-ring into the housing (if used) • Reassemble housing to the system head • Restore water supply to the system • Flush the system as recommended • Check housing seal for leaks • Record the installed and change date Estimated activity time: 15 minutes 在这个活动中，学员练习更换置入式水处理系统的滤芯。讲师必需透过列的任务解释、示范并教导学员： • 将系统的水供给截断 • 释出系统压力 • 将滤水器外壳自滤头上松开 • 将用过的滤芯及 O 环移除 • 清洁并消毒滤水器外壳 • 将新滤芯装进滤水器外壳 • 润滑并将 O 环放回滤水器外壳 • 将滤水器外壳锁回系统滤头 • 恢复系统水供给 • 按建议以水冲洗系统 • 确认滤水器外壳锁紧无泄露 • 纪录此次滤芯更换日期及下次预计更换日期

	预估活动时间:15 分钟		
1.08 PM Components 预防性维护零件	In this activity, learners interact with and learn to identify physical components from espresso machines, coffee batch brewers, and coffee grinders which are often recommended to be maintained or inspected during a Preventive Maintenance service. Components shall be displayed within a demonstration batch brewer, espresso machine, and espresso grinder, with all equipment disconnected from power supply. Additional samples of components should also be displayed individually outside of the equipment. Ensure that the students interact with the equipment which will be used in Activity 1.13 and the Practical Exam. This can be done throughout the course or at one specific activity. 在这个活动中，学员将会学习辨认预防性维护服务中一般建议维护或检查的义式咖啡机、滴滤式咖啡机与咖啡磨豆机的实体零件并与之互动。零件在展示时应该在示范的滴滤式咖啡机、义式咖啡机及义式咖啡磨豆机内，所有设备皆不得接上电源。额外的零件样本也应该个别地在设备外展示。确保学生与活动 1.13 及实务测验中所使用设备有所互动，这可以透过课程或一个指定的活动来达成 The components to be identified include 应被辨识的零件包含:		
	Espresso machine brewing group: 义式咖啡机冲煮头: - Group head 冲煮头 - Brew valve* 冲煮阀 - Portafilter locking bezel * 冲煮把手固定环 - Group gasket 冲煮头垫圈 - Dispersion disc* 分水块 - Group screen 冲煮头分水网 - Group screw* 分水网螺丝 - Portafilter 冲煮把手 - Portafilter basket 冲煮把手滤杯 - Portafilter basket retaining spring 滤杯固定弹簧	Espresso machine steam valve: 义式咖啡机蒸气阀 - Valve body 阀门本体 - Valve stem 蒸气阀顶杆 - Valve seal 阀体垫片 - Valve o-rings 阀门 O 环 - Valve spring 阀门弹簧 - Valve handle 阀体扳杆	Coffee grinder: 咖啡磨豆机 - Bean hopper 储豆槽 - Grinding chamber 研磨腔室 - Grinding burrs 刀盘 - Burr carrier 刀座 - Burr carrier shear plate* 刀座插销 - Outlet chute 出粉通道 - Declumping device* 防结块装置 Note: show components representing both espresso and filter coffee grinders 注意: 展示零件应能够代表义式及一般单品磨豆机
	Espresso machine steam boiler:	Ball-end style espresso machine steam	Batch brewer:

	义式咖啡机蒸气锅炉: - Pressure safety valve 压力安全阀 - Anti-vacuum valve 真空排除阀 - Water level probe 水位探针 - Pressure switch* 压力开关 wand joint assembly: 义式咖啡机球型接头蒸气棒 - Steam joint nut 蒸汽管固定接头螺帽 - Steam wand ball-end 蒸气棒球型接头 - Spring 弹簧 - Spring bushing 弹簧衬套 - O-ring O 环 - Gasket 垫片 滴滤式咖啡机 - Water level probe 水位探针 - Spray head 喷洒头 *Note that equipment designs vary and not all equipment will exhibit all components indicated. Components not presented on demonstration equipment may be represented by pictures. * 注意设备的设计是多样的，并非所有的设备都会有标示的所有零件。未在示范设备中展示的零件需以照片形式来说明。 Estimated activity time: 30 minutes 预估活动时间:30 分钟
1.11.02 PM Checklist 预防性维护检核表	
1.11.03 Backflushing an Espresso Machine Brew Group 义式咖啡机冲煮头的逆洗	In this activity, the Trainer demonstrates steps recommended for backflushing an espresso machine brew group using detergent. 在这个活动中,讲师示范使用清洁剂逆洗义式咖啡机冲煮炉头的建议步骤 Estimated activity time: 5 minutes 预估活动时间:5 分钟
1.11.04 - 1.11.05 Espresso Machine PM 义式咖啡机的预防性维护	In this activity, learners practice recommended maintenance and inspection steps for espresso machine PM service. Ensure that the Learners interact with the same or identical espresso machine that will be used in the Practical Exam. Trainer must explain, demonstrate, and guide learners through the following tasks: ● Replace group screen and gasket ● Rebuild a ball-end style steam wand joint ● Inspect an espresso machine for proper function ● Record service activities on a PM checklist 在这个活动中,学员练习义式咖啡机预防性维护服务中建议的维护及检查。确认学员与实务测验中所使用同型或同一台的义式咖啡机有所互动。讲师必需透过以下任务解释、示范及教导学员:

	• 更换炉头分水网及垫圈 • 更换球型蒸气棒接头 • 检查义式咖啡机的正常功能运作 • 在预防性维护检核表上纪录下服务活动 Estimated activity time: 45 minutes 预估活动时间:45 分钟
1.11.06 Batch Brewer PM 滴滤式咖啡机的预防性维护	In this activity, learners practice inspecting and cleaning a batch brewer fill level probe. Ensure that the Learners interact with the same or identical batch brewer that will be used in the Practical Exam. Trainer must explain, demonstrate, and guide learners through the following tasks: • Confirm the brewer is disconnected from electrical power and remove necessary panels • Remove, inspect, and clean level probe • Reinstall level probe and reassemble panels 在这个活动中，学员练习检查并清洁滴滤式咖啡机的水位探针。确认学员与实务测验中所使用同型或同一台的义式咖啡机有所互动。讲师必需透过以下任务解释、示范及教导学员： • 确认滴滤式咖啡机并无接上电源并移除必要的隔板 • 拆除、检查并清洁水位探针 • 重新装上水位探针并将隔板重新组装恢复 Estimated activity time: 15 minutes 预估活动时间:15 分钟
1.11.07 Espresso Grinder Preventive Maintenance 义式咖啡磨豆机的预防性维护	In this activity, learners practice replacing espresso grinder burrs. Ensure that the Learners interact with the same or identical espresso grinder that will be used in the Practical Exam. Trainer must explain, demonstrate, and guide learners through the following tasks: 在这个活动，学员练习更换义式咖啡磨豆机的刀盘。确认学员与实务测验中所使用同型或同一台的义式咖啡磨豆机有所互动。讲师必需透过以下任务解释、示范及教导学员： • Disconnect grinder from electrical power • Open grinding chamber and removes worn burrs • Thoroughly clean grinding chamber and burr carriers • Install new grinding burrs using an appropriate screw tightening procedure • Reassemble grinder • Calibrates grinder and adjusts to an appropriate setting • 断开磨豆机的电源 • 打开研磨腔室并移除旧的磨损刀盘 • 彻底清洁研磨腔室及刀座 • 换上新的刀盘并以正确地程序栓紧螺丝 • 重新组装恢复磨豆机 • 校正磨豆机并调整至适当的设定 Estimated activity time: 30 minutes

	预估活动时间:30 分钟
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Appendix B: Glossary of Terms

附录 B: 专业术语

Course Term 课程专业名词	Definition or Description 定义或描述	Alternative Terms 替代术语	Reference 相关资源
Alkalinity 碱度	The ability of the water to neutralize acids. Also describes the concentration of carbonate and bicarbonate compounds in the water. 水能够中和酸的能力。也描述水中碳酸根及碳酸氢根化合物的浓度		SCA Water Quality Handbook, pg. 77
Anion 阴离子	Negatively charged ion. 带负电的离子		SCA Water Quality Handbook, pg. 77
Anti-vacuum valve 真空排除阀	A mechanical component that allows air to enter a steam boiler or circuit to break a vacuum that may occur as the chamber containing the steam cools. 当锅炉中内含蒸气冷掉后，让空气进入蒸气锅炉或破坏真空的机械性零件	Vacuum breaker 真空泄压阀	
Backflush 逆洗	A procedure for rinsing or cleaning espresso machine brewing groups by causing water or detergent solution to flow backwards from a blank portafilter basket, through the shower screen and brewing valve, and out through the brew valve exhaust port. 藉由让水或清洁溶剂从无孔滤杯回流经分水网、冲煮阀而由冲煮阀排水口而出的方式，达到冲洗或清洁义式咖啡机冲煮头的程序		
Ball-end joint 球型接头	A pivoting joint, often found on the end of a steam or hot water wand, consisting of a rotating spherical surface which is pressed against a concave metal surface or polymer gasket by a spring and spherical bushing to create a leak-proof seal. 一个旋转接头，通常在蒸气棒或热水棒的末端，由一个旋转球型面透过一个弹簧及球型衬套压紧在金属凹面或聚合物垫片上，创造出防漏的密封状态		
Bar 大气压力	A unit of measurement for pressure as related to the atmosphere. One bar is Equal to (1) atmosphere or 14.7PSI(G) 一个与大气有关测量压力的单位。一个 bar 等于一大气压力或 14.7PSI(G)		
Bean hopper 咖啡豆槽	A coffee grinder component that holds coffee beans to be ground 将待磨咖啡豆托存住的一个咖啡磨豆机零件		

Boiler level sight glass 锅炉水位窗口	A glass component that allows the user to see a representation of the current water level in a steam boiler. 让使用者可以看到蒸气锅炉中目前水位表示的一个玻璃零件		
Boiler safety valve 锅炉安全阀门	A safety component mounted to a steam boiler that relieves excess pressure. 一个安装在蒸气锅炉的安全零件，可以释放出多余的压力		
Brew pressure gauge 冲煮压力表	An instrument that measures and displays the pressure of the water being used for espresso extraction. 测量并显示义式咖啡萃取时水压的仪表		
Brew valve 冲煮阀	An electro-mechanical valve that opens to allow the flow of water to the coffee bed when a brew cycle is activated. 一个电子机械阀，当冲煮循环启动时，打开时让水流经咖啡粉饼	Three-way valve 三通阀	
Burr carrier 刀座	A coffee grinder component used to mount a grinding burr. Grinders typically contain one stationary burr carrier and one moving burr carrier, often mounted to the output shaft of a motor. 一个安装于磨豆机刀盘的咖啡磨豆机零件。磨豆机通常包含一个固定的刀座和一个移动的刀座，通常安装在马达的输出轴上		
Burr carrier shear plate 刀座插销	A coffee grinder component which transmits torque from the grinder motor output shaft to the moving burr carrier under normal use but is designed to fail if excessive loads are encountered in order to minimize damage to the grinder. 一个咖啡磨豆机零件，在正常使用下，由磨豆机马达输出轴传递扭力至移动的刀座，但为了减少对磨豆机的损害，在多余的负载发生时失去功效		
Carbonate Hardness 碳酸盐硬度	Carbonate hardness effectively corresponds to the maximum amount of scale that can form and is therefore equal to whichever of the two values of total hardness and alkalinity is lower. Synonymous with Temporary Hardness. 碳酸盐硬度实际上相当于结垢的最大量，因此等同于总硬度及碱度二者中比较低的，为暂时硬度的同义词	Temporary Hardness 暂时性硬度	SCA Water Quality Handbook, pg. 77
Cation 阳离子	Positively charged ion. 带正电的离子		

Chloride 氯离子	The chloride ion is the anion Cl ⁻ . It is formed when the element chlorine (a halogen) gains an electron or when a compound such as hydrogen chloride is dissolved in water. 氯离子即是阴离子 Cl ⁻ ，它的形成来自于氯元素得到了一个电子或当像氯化氢的化合物溶解于水中时		Chloride - Wikipedia
Chlorine 氯	Tap water is chlorinated with hypochlorite (Cl ₂) to kill bacteria and microbes. Not to be confused with chlorides such as sodium chloride (NaCl), calcium chloride (CaCl ₂), or magnesium chloride (MgCl ₂). 自来水使用次氯酸盐的氯化来消灭细菌及微生物，不要与氯化钠(NaCl)、氯化钙(CaCl ₂)或氯化镁(MgCl ₂)中的氯离子混淆		SCA Water Quality Handbook, pg. 78
Compound 化合物	An electrically neutral group of two or more atoms of the same or different elements, held together by chemical bonds (e.g. Oxygen (O ₂); Water - two hydrogen atoms bonded to an oxygen atom (H ₂ O); or Salt (NaCl). 一个有着二个或多个一样或不同元素的不带电的原子团，透过化学键绑在一起 (例:氧(O ₂)、水-二个氢原子键合一个氧原子(H ₂ O)或盐(NaCl).)		SCA Water Quality Handbook, pg. 78
Decarbonizer 脱碳剂	Decarbonization is also an ion-exchange method, though in contrast to the former, the calcium and magnesium ions are exchanged for protons (H ⁺). The released protons in turn neutralize hydrogen carbonate by formation of carbonic acid. Decarbonization leads to a decrease in total hardness and alkalinity by equal amounts. 脱碳剂也是一种离子交换的方法，与之前相比较，钙离子及镁离子与氢离子(H ⁺)交换。释放出的质子反过来中和碳酸氢盐形成碳酸。脱碳使总硬度及碱度以相同量值的减少。	Decarbonizing Softener, alkalinity Softener 脱碳软化, 去碱度软化	SCA Water Quality Handbook, pg. 78
Declumping device 防结块装置	A coffee grinder component which breaks up clumps of coffee grounds in the grinder output chute. 咖啡磨豆机中的装置，在磨豆机的出粉通道中破除咖啡粉的结块		
Dispersion Plate 分水块	The part of the group head assembly that distributes water in order to shower over the dispersion screen. This is also referred to as a shower plate. 咖啡冲煮头中一部分的组合物，将水以花洒状分配到分水网上，也被称为花洒盘	Dispersion disc 分水片	
Drain tube 引流管	Tubing dedicated to connecting the drain manifold to the drain system of a facility or a dedicated receptacle.		

	专用来连接排水歧管至某一个装置的排水系统或专门容器的管道		
Electrical Conductivity 导电度	Electrical conductivity (EC) measures the ability of a substance to transmit or conduct electricity. Since pure water is a pure conductor, EC in drinking water originates mainly from dissolved ions. Electrical conductivity has often been used to estimate the total dissolved solids content (TDS) of water samples but conversion factors can be up to +/- 50%. 导电度测量某一物质传递或导电的能力。因为纯水是几乎不导电的，饮用水中的导电度主要来自于溶解的离子。导电度也常用来估测水样本的总溶解固体含量 (TDS)，但转换因子至多可到达+/-50%。		SCA Water Quality Handbook, pg. 78
Expansion valve 膨胀阀	A component that relieves pressure from brew boilers at a set point, which is often 12 bars. May or may not be field-adjustable. 这个零件在来自冲煮锅炉的压力达到一个设定点，通常是 12 个 Bar 时会释放压力。可能也可能不能调整范围		
Flowmeter 流量计	A measurement device that tracks the volume of liquid passing through it and sends information to a control board. 追踪通过它的液体体积并传送讯息给控制板的量测设施		
Grains per Imperial Gallon 格令/英制加仑	A unit of water hardness defined as 1 grain (64.8 milligrams) of calcium carbonate dissolved in 1 US gallon of water (3.78 L). It translates into 1 part in about 58,000 parts of water or 17.1 parts per million (ppm). Also called Degrees Clark. 一个水的硬度单位，定义为 1 格令(64.8 毫克)的碳酸钙溶解在 1 美式加仑的水 (3.78L)。它可以转化为 1 等份对大约 58000 等份的水或 17.1 等份每百万(ppm)。也称为克拉克硬度	Degrees Clark 克拉克硬度	SCA Water Quality Handbook, pg. 78
Grinding burr 研磨刀盘	A coffee grinder component, typically made from hardened steel or ceramic material, and featuring a large number of teeth which are used to fracture coffee beans into progressively smaller particles. 咖啡磨豆机的零件，通常由硬化钢或陶瓷材料所制成，特质上有着大量用来使咖啡豆进一步破碎成更小的颗粒的牙	Grinding blade	
Grinding chamber	An enclosure within a coffee grinder which contains the grinding burrs, directs the flow		

研磨腔室	of whole coffee beans into the burrs, and directs the flow of ground coffee particles into an outlet chute. 咖啡磨豆机内一个封闭区域，内含研磨刀盘、将整颗咖啡豆送至刀盘，并将研磨后的咖啡颗粒送进出粉通道。		
Group/group Head 冲煮头	An assembly of components that receives the portafilter and delivers brewing water to the coffee bed at a preset temperature and flow rate, AKA brew unit/pouring unit. 接上把手并将冲煮水以预定的温度及流速送到咖啡粉饼上的零件组合体，也称为冲煮组件	Brewing group 冲煮头	
Hardness 硬度	Now referred to as 'total hardness', this is the concentration of calcium (Ca) and magnesium (Mg) in the water. More calcium and magnesium = harder water. Less calcium and magnesium = softer water. 现在称为总硬度，这是水中钙与镁的浓度。越多的钙与镁=较硬的水。越少的钙与镁=较软的水		SCA Water Quality Handbook, pg. 79
Ion 离子	Atoms or molecules that have a net positive or negative charge. 有着净正电荷或净负电荷的原子或分子		SCA Water Quality Handbook, pg. 79
Langelier Scaling Index 朗格利尔饱和指数	An approximate indicator of the degree of saturation of calcium carbonate in water. It is calculated using the pH, alkalinity, calcium concentration, total dissolved solids, and water temperature of a water sample collected at the tap. 水中碳酸钙饱和度的一個大約指标。它的计算使用自来水水样本的 pH、碱度及钙浓度，总溶解固体含量和水温		SCA Water Quality Handbook, pg. 79
Larson-Skold-Index 拉森指数	A measure for the corrosivity of water towards steel. It is calculated by dividing the sum of chloride and sulfate by alkalinity. 水对钢材侵蚀性的量测，计算使用氯及硫酸盐的加总除以碱度		SCA Water Quality Handbook, pg. 79
Limescale 结垢	The hard white chalky deposit found in kettles and boilers. It originates from limestone that is dissolved from the ground and then re-forms later in equipment when the water chemistry changes. It is mostly calcium carbonate (CaCO ₃), but in many places magnesium carbonate (MgCO ₃) may account for up to 1/3 of the total. 在水壶或锅炉中所找到白色硬粉状的沉淀物，来自于地层中石灰岩的溶解物，然后因水质的化学改变，在设备中再次成形。它多数来自于碳酸钙(CaCO ₃)，但在很		Equilibrium in Water and Coffee, pg. 32

	多地方，碳酸镁可能至多会占其中的 1/3		
Locking bezel 固定框架	Ring-shaped portion of the brewing group with tapered slots which accept the tabs on the sides of a portafilter and are designed to press the basket rim against the group gasket when locked into place to create a seal. 环形部分的冲煮头，有着变窄的狭槽能够让冲煮把手二侧的襟片卡上，并被设计为将滤杯的边缘压在冲煮垫片上而锁进位置上创造出密封状态		
Maintenance interval 保养周期	A recommendation for the amount of time a component or system should be used between preventive maintenance service activities. 建议在预防性维护服务之间一个零件或系统的使用寿命		
mg/L 毫克/升	Milligrams per liter. It is the equivalent to ppm when testing water at <45°C. 毫克每升。当测试温度<45°C，它等同于 ppm		SCA Water Quality Handbook, pg. 79
Molecule 分子	An electrically neutral group of two or more atoms held together by chemical bonds. For example, Oxygen (O ₂) or Water (H ₂ O). 一个有着二个或多个透过化学键绑在一起的不带电原子团 例如: 氧(O ₂)或水(H ₂ O).		SCA Water Quality Handbook, pg. 79
Non-Carbonate Hardness 非碳酸盐硬度	The concentration of calcium and magnesium occurring as compounds other than carbonates in the water. 钙镁在水中以非碳酸盐化合物存在的浓度		SCA Water Quality Handbook, pg. 79
Not Hard Carbonates 非硬度碳酸根	The concentration of minerals, other than calcium and magnesium, which occur as carbonate compounds. 非钙镁的矿物质以碳酸盐化合物存在的浓度		SCA Water Quality Handbook, pg. 79
Outlet chute 出粉通道	A feature of a coffee grinder which directs the flow of coffee grounds from the grinding chamber into the desired location, typically a portafilter basket, ground coffee funnel, brew basket, coffee bag, or dosing cup. 咖啡磨豆机中引导咖啡粉由咖啡腔室进入预期的位置，通常是冲煮把手的滤杯、咖啡粉料斗、冲煮滤杯、豆袋或接粉杯中		

Permanent Hardness 永久硬度	The calcium and magnesium that cannot precipitate from water when boiled. 当水滚时无法由水中沉淀而出的钙与镁		SCA Water Quality Handbook, pg. 79
pH	This describes whether the water is more acid, more alkaline, or neutral. 这描述了水是否是较酸、较碱或中性		SCA Water Quality Handbook, pg. 79
Polyphosphate 多磷酸盐	Food grade polyphosphates, in basic phosphorus bonds, react with the lime minerals in the water and thus disturb the scale deposits. 食品等级的多磷酸盐，以基本的磷键结构，与水中的水垢矿物质产生反应，因此扰乱了水垢的沉淀	Scale Inhibitor 阻垢剂	Equilibrium in Water and Coffee, pg. 71
Portafilter 冲煮把手	A removable, handheld component of a lever, semi-automatic, or volumetric espresso machine that performs the function of retaining the filter basket and securing it into the group head during the extraction process. It includes a pouring spout to direct the flow of coffee into a container. 一个可移除、手持式的零件，在拉把式义式咖啡机、半自动义式咖啡机或定量式义式咖啡机萃取的过程中，容纳滤杯并将其紧扣至冲煮头中。它包含了将咖啡液导流至容器中的导流嘴。		
Portafilter Basket 冲煮把手滤杯	A component of a traditional espresso machines that serves as the container for the ground coffee during extraction. The bottom of the basket features finely stamped or machined holes that allow the extracted coffee to exit while retaining the grounds. 传统义式咖啡机中在萃取时作为咖啡粉容器的一个零件。滤杯的底部有着许多细小的机器冲孔，让萃取后的咖啡可以流出却拦住咖啡粉		
Ppm	Parts per million. It is equivalent to mg/L when testing water at <45°C. 百万分之一，当测试温度<45°C，它等同于克/毫升		SCA Water Quality Handbook, pg. 79
Pressure gauge 压力表	An instrument that measures and displays the current pressure within the steam or coffee boiler on an espresso machine. 一种设备，可以测量并显示义式咖啡机内的蒸气或咖啡锅炉的最近温度		
Pressure switch 压力开关	An electrical switch that is activated by a pressure sensitive mechanism to control the power supply to an espresso machine steam boiler heating element. 一种电子开关，由压力敏感机制所触发，进而控制义式咖啡机蒸气锅炉加热组件	Pressure stat 自动调压器	

	的电力供给		
Preventive Maintenance 预防性维护	Regular and routine service performed to maintain proper system function and reduce unexpected equipment failure. 定期常规的服务执行以维持适当的系统功能并减少无法预期的设备故障		
PSI	A unit of measurement for pressure, representing Pounds per Square Inch 一种测量压力的单位，代表每一平方英寸的磅数		
Pump pressure 泵浦压力	The pressure of a liquid when water is moving through the system. 当水正在穿过系统时的液体压力		
Reactive Maintenance 反应性维护	Unplanned repair of equipment that has failed in order to restore proper system function. 设备故障后,为了让系统回到正常功能运作而进行未在计划内的修复	Emergency service 紧急维修	
Reverse Osmosis 反渗透	A pressure-driven membrane process used for purification of water. 一种压力趋动穿过滤膜的过程，用以进行水的纯化		Equilibrium in Water and Coffee, pg. 92
Shower screen 分水网	A fine, wire-mesh screen that evenly distributes brew water as it flows into a portafilter. This is also referred to as a dispersion screen. 一个细密的网状滤网，让冲煮水得以平均分布地流进把手，英文也称之为 dispersion screen。	Group screen 冲煮头分水网	
Shower screw 分水网螺丝	Screw, usually made of stainless steel, which secures the shower screen to the group 将分水网栓进冲煮头的螺丝，通常为不锈钢制	Group screw 冲煮头螺丝	
Shutoff valve 关闭阀	An inline component that is used to control the flow of water or steam manually. 一个串联零件，用来手动控制水或压力的流动		
Sodium (Na) 钠	Alkali metal that can occur as a chloride compound (e.g. Salt (NaCl)), or as a carbonate compound in water. 碱金属，可以以氯化物(例: 盐(氯化钠))或以一种碳酸盐化合物存在于水中		SCA Water Quality Handbook, pg. 80

Softener 软化器	A softener is an ion-exchange method where calcium and magnesium ions are exchanged for either sodium or potassium ions. This treatment reduces total hardness without affecting alkalinity. 软化器是一种离子交换的方式，用钠离子或钾离子来交换钙离子和镁离子，这样的处理在不影响碱度的状态下减少了总硬度		SCA Water Quality Handbook, pg. 80
Spray head 喷洒头	Batch brewer component that promotes the uniform distribution of water onto the bed of coffee grounds 滴滤式咖啡机中的零件，促使水均匀的分布进咖啡粉饼中		
Steam boiler 蒸气锅炉	A vessel where water is converted from liquid to steam and stored under pressure. 水在这个容器中由液体转换成蒸气并在压力储存其中		
Steam valve 蒸气阀	A component on espresso machines that controls the flow of steam from the boiler to the steam wand, either by a control lever, knob, or electro-mechanical solenoid and switch. 义式咖啡机中的零件，控制蒸气的流动由锅炉流向蒸气棒，或由控制杆、旋钮或机电螺线管及开关		
Steam wand 蒸气棒	A teflon, stainless or plated brass component used for steaming milk, with a removable tip that releases a controlled blast of steam through one or more small holes. 用来蒸牛奶的一种铁氟龙、不锈钢或镀黄铜的零件，有着可移除的尖端，可透过一个或多个小洞释放出一股可控制的蒸气		
Supply line 供水线	A tube or hose connecting the building's water supply to the machine's inlet/fill valve. 连接建筑物供水至机器进水阀的软管或金属管		
Temporary Hardness 暂时硬度	The calcium and/or magnesium can precipitate from the water when boiled, and form scale deposits. 当水煮滚后,会从水中沉淀析出的钙和镁，并形成结垢沉积物		SCA Water Quality Handbook, pg. 80
Total Dissolved Solids (TDS) 总溶解固体含量	Total dissolved solids simply means the sum of all solid substances present in the water. For water, this parameter is often estimated by electrical conductivity by assuming a fixed composition. A conversion factor of 0.7 in mg/L of TDS per 1 µS/cm of electrical		SCA Water Quality Handbook, pg. 81

	conductivity is used by the heritage SCAA Water Standard. In laboratory tests, it is measured by evaporation and weighing of the solid residue – also including uncharged dissolved solids such as silicates that do not contribute to electrical conductivity. 总溶解固体仅表示水中所有存在的固体物质总和。对水来说，在假设组成固定的状态下，这个参数通常会以导电度来作估测。换算因子以每 1 $\mu\text{S}/\text{cm}$ 的导电度，则有 0.7 毫克/升的总溶解固体，继承自 SCAA 水质标准的遗产 在实验室的测试中，它的量测会透过蒸发并对固体残留物进行称重，这也包含了不导电的溶解固体，像是硅酸盐，它们并无法贡献导电度		
Total Hardness 总硬度	Defined as the sum of calcium and magnesium in equivalent concentrations (or molar concentrations). In rare cases, other ions can contribute to total hardness, for example strontium. 定义为在当量浓度(莫耳浓度)下，钙与镁的总和，在极少的案例中，其他离子亦对总硬度有所贡献，例如像锶		SCA Water Quality Handbook, pg. 81
Water pump 水泵	A positive-displacement mechanism that traps the liquid entering from an inlet pipe and forces it out through a discharge pipe. It is operated by an attached motor. 一个正排气型的机制，把液体从进水管中收集起来，而后迫使其由排放管中倾泻而出。藉由一台附加的马达而达成运作		
Water-level fill probe 满水位探针	An electrical conductivity sensor that signals for a boiler to be filled to a desired level. 电子导电度传感器会在锅炉将满到预期水位时发出信号		